

**IOT BASED SMART LANGUAGE MONITORING SYSTEM
FOR PRIMARY SCHOOLS**

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**A report submitted in partial fulfilment of the requirements for the
NMJ32004 Integrated Design Project course**

**FACULTY OF ELECTRONIC ENGINEERING TECHNOLOGY
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JULY 2025

DECLARATION SHEET

We declare that the report entitled “IOT BASED SMART LANGUAGE MONITORING SYSTEM FOR PRIMARY SCHOOLS” and the work presented in the report are both our own, and have been generated by us as the result of our original design project. We confirm that this report is made to appropriate standard in terms of scope, quality and presentation as partial fulfilment of the requirement for the Design Project Course in Universiti Malaysia Perlis (UniMAP).

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LIST OF ABBREVIATIONS

IoT	Internet of Things
IDE	Integrated Development Environment
ESP	Embedded System
I2C / I ² C	Inter-Integrated Circuit
OLED	Organic Light-Emitting Diode
SDA	Serial Data
SCL	Serial Clock
I2S / I ² S	Inter-IC Sound
SDG	Sustainable Development Goals
AI	Artificial Intelligence

Chapter 1

Introduction

1.1 Project Background

Language skills are one of the most important aspects in education, especially in country like Malaysia. Based on an article from Open University, language skills empower learners to understand deeply, communicate effectively and navigate their professional and personal growth [1]. However, in most primary schools, monitoring language use among students is still done manually by teachers. This method is not only time-consuming but also makes it difficult to give attention every student. With large classroom sizes, teachers often face challenges in speaking and communication abilities of their students.

With the advancement of technology and the concept of smart classrooms, it is now possible to use IoT devices to help improve learning environments. For this reason, our group proposed an IoT-Based Smart Language Monitoring System that aims to support students in practicing different languages in a more interactive way, especially during their daily routine at the canteen. By using basic IoT components and voice recognition technology, this system allows students to speak in a set language of the day and receive instant feedback. The system also logs their performance data into a cloud platform, which helps teachers monitor their progress more easily and efficiently.

1.2 Description of the Problem.

This project idea came after observing that most students tend to communicate in their own language throughout the day. To validate this observation,

we conducted an online survey to see what is their daily language usage. We asked a few questions such as:

- What is your mother tongue?
- What language do you speak most often at home?
- What language do you use most frequently with friends?
- What language do you use when ordering food?
- Do you find yourself mixing languages when ordering food?

From the results based on the survey, we concluded that a majority of students use English and Malay when ordering food. We also concluded that a majority of students mix their languages when ordering food. This shows that students prefer to order in the main two languages, Malay and English [2]. Even though English, Chinese, or Tamil are taught in the classroom, there is not much opportunity for students to practice them outside formal lessons. At the same time, teachers mentioned that they do not have enough time to assess each student's speaking ability individually.

Plenty of articles from education websites also supported the need for digital learning tools that can assist with this kind of assessment. For example, based on an article from Language Tech Lab, this website lists a few tools that can be used for language proficiency modes that can suffice the use of different languages [3]. Another example is an article from Edutopia, which states that languages are assets and as a way to build new vocabulary [4]. Based on all these inputs, we realized there is a real need for a smart, automated language monitoring system that works in everyday school settings like the canteen.

1.2.1 Identification of the problem...

In primary school education, the development of language skills such as speaking, reading, and writing is fundamental to a child's academic and social success. However, many schools, especially those in large classrooms or under-

resourced environments, face significant challenges in effectively monitoring and supporting students' language development on an individual basis. These challenges are also faced by the educators teaching these young minds.

Speaking of educators, teachers often rely on periodic assessments, manual observations, or subjective evaluations to track student progress in language. These methods are time-consuming, inconsistent, and not scalable, especially in classrooms with a high student-to-teacher ratio. Teachers are burdened with multiple responsibilities and administrative tasks, leaving limited time for personalized language assessment for each student.

Next, traditional language assessments provide delayed feedback, which reduces the ability to intervene early when students show signs of difficulty in speech clarity, vocabulary use, or comprehension. Furthermore, early signs of language delays or speech disorders often go unnoticed until later stages, especially when students are quiet or do not actively participate in class. This can result in long-term academic struggles if not addressed early.

Finally, while schools are increasingly adopting digital tools for teaching, few have implemented smart systems to actively monitor and analyse students' language use during classroom interactions. These problems indicate a clear need for an automated, intelligent system which includes the Internet of Things framework which provide convenience that can assist educators in continuously and objectively monitoring student language development in real time thus resulting in the improvement of proper language use in the young minds of students in primary school.

1.2.2 Proposed solution for the problem

To address this issue, our team designed an IoT-based system that is installed at the school canteen. It uses a microphone and voice recognition to detect the language spoken by the student when they order food. If the student uses the correct

language of the day, they will receive positive feedback through a display and speaker. If the wrong language is used, the system will prompt them to try again in the correct one. All this information is recorded in a Microsoft Excel sheet, which is only accessible by teachers. This way, students get more chances to speak and practice, while teachers can monitor performance without having to manually observe every interaction.

1.3 Project Objectives

The project is aimed to:

- a) To develop an IoT-based system that allows primary school students to practice speaking different languages through daily activities.
- b) To integrate voice recognition technology to detect spoken language and provide real-time feedback.
- c) To assist teachers in monitoring student language use by automatically logging speech data into a cloud platform.

1.4 Project Scope

The system is designed to detect four languages Malay, English, Chinese, and Tamil and is specifically intended to be used in a school canteen setting. It uses an ESP32 microcontroller connected to a microphone, speaker, and OLED display. The speech data is recorded and uploaded to Microsoft Excel Sheets for teacher review. However, this system is limited to basic language detection only and does not include grammar correction. It also relies on internet connectivity to function properly and is not suitable for offline use at this stage. Additionally, the system is designed for use in a primary school cafeteria environment, considering child safety, simplicity, and ease of use.

Chapter 2

Methodology

2.1 Introduction

The methodology adopted in this project outlines the systematic steps taken in the design, development, and testing of the IoT-Based Smart Language Monitoring System. This system integrates hardware and software components. To achieve a functional prototype that supports language practice and feedback in a primary school canteen environment. The methodology includes project design strategies, schematic and circuit design, 3D modelling for casing, physical prototyping, and comprehensive testing procedures. Each stage was developed with careful consideration of technical feasibility, usability, and ethical responsibility, ensuring that the final system aligns with the project objectives and user requirements.

2.2 Project Design

The development of the system followed a modular and iterative approach. The design was divided into key phases, starting with component selection and schematic design, followed by hardware integration, 3D enclosure modelling, and final prototype construction. The system is composed of an ESP32 microcontroller, OLED display, microphone, MAX98357A audio module & speaker and a two button keyboard interface. A cloud-based data logging mechanism was implemented using PHPMyAdmin database, and a teacher dashboard was developed using HTML, PHP,

and MySQL. All the data that is recorded from the system is exported into a Microsoft Excel sheet for the future use of teachers in implementing different language usage when ordering food in a primary school canteen setting.

2.2.1 Hardware Selection and Justification

The hardware selection was based on availability, reliability, compatibility with the ESP32 platform, and suitability for educational environments. Cost-effectiveness and ease of integration were also considered.

- **ESP32 Microcontroller**



Figure 2-1 ESP32 DEV Module

This ESP32 module acts as the brain of the circuit by providing the function to the speaker and OLED screen which will explained later on. This chip provides plenty of features such as Wi-Fi and Bluetooth compatibility for embedded devices [5]. This chip is most commonly used for mobile devices, wearable tech and amateur IoT projects. The price for one of this module is quite reasonable and affordable for this project. In order to operate this module, a special type of software is needed as shown in Figure 2-2:

```
Sam_ESP | Arduino 1.8.20 Hourly Build 2022/04/25 09:33
File Edit Sketch Tools Help

Sam_ESP | good_job.h | wrong_answer.h

#include <WiFi.h>
#include <WebServer.h>
#include <Adafruit_GFX.h>
#include <Adafruit_SSD1306.h>
#include <driver/t2a.h>
#include <good_job.h>
#include <wrong_answer.h>
#include <pgmspace.h>

// ===== WiFi Configuration =====
const char* ssid = "SamWiFi"; // Replace with your WiFi name
const char* password = "Lampay2302"; // Replace with your WiFi password

// ===== OLED Setup =====
#define SCREEN_WIDTH 128
#define SCREEN_HEIGHT 64
#define OLED_RESET -1
Adafruit_SSD1306 display(SCREEN_WIDTH, SCREEN_HEIGHT, &Wire, OLED_RESET);

// ===== I2S Speaker Pin Config =====
#define I2S_BFF_DATA 33
#define I2S_BFF_CLOCK 27
#define I2S_BFF_LATCH 26

// ===== Audio Config =====
#define SAMPLE_RATE 16000
#define BUFFER_SIZE (SAMPLE_RATE * 2)
int16_t audioBuffer[BUFFER_SIZE];

// ===== Web Server =====
WebServer server(80);

// ===== OLED Display Function =====
void showOnOLED(String line1, String line2 = "") {
  display.clearDisplay();
```

Figure 2-2 Arduino IDE software

This software is known as the Arduino software (IDE) which uses a text editor such as Microsoft Visual Studio code which communicates with boards such as ESP32, Arduino Uno and Raspberry PI [6]. The following code that is used in this project is available in Appendix B.

- **SSD1306 0.96 inch I2C OLED display**



Figure 2-3 OLED Display

This display as shown in Figure 2-3 plays a vital role in our system as it notifies the students whether they are speaking the targeted language of the day and giving them transcription of what is actually being spoken. The I2C in the name represents the the serial communication protocol that only uses two wires, SDA (Serial Data) and SCL (Serial Clock). This protocol allows it to communicate with each other by only using two wires and unique addresses [7]. This specific display is used because is OLED is well-known for its low power consumption and brightness [8]

- **Microphone**



Figure 2-4 TableTop Microphone

This microphone as shown in the figure above was used to improve the clarity of the user as the original microphone on the laptop could not be amplified and

produced noisy results when playback. Initially, the INMP441 module as shown in Figure 2-5 was used in the system but has since been removed due to the audio capture quality being horrendous and background noise was amplified which made the speech recognition process much harder and sometimes unrecognizable.



Figure 2-5 INMP441 Microphone Module

- **MAX98357A audio module & speaker**



Figure 2-6 Audio Module & Speaker

This audio module is known as the easiest and greatest digital audio amplifier breakout board. It takes standard I2S digital audio input and not only decodes into analog but amplifies it directly into a speaker [9] I2S is another protocol that is mainly used for sound and is implemented on circuits boards such as ESP32 to pass audio data around. The speaker used is a 3W Yamaha speaker which produces loud and good quality sound for its price. This speaker is used to notify the students via voice whether they are correct or wrong in speaking the targeted language.

- **Two Button Keyboard Interface**



Figure 2-7 Button interface

This type of interface was recommended as it provides simplicity and preciseness in the system. It uses a USB interface which will be connected to the PC and the two buttons are programmable shortcuts that will run the system automatically without any issue. Initially, a small single push button was suggested, however during the creation of the prototype, it became obvious that it was too small and would not fit the the location (which is a primary school canteen). The two buttons can be programmed using a software to run different keyboard shortcuts as well as run the processing script for the speech recognition section. Overall, this two button keyboard made the system much more easier to interact.

2.2.2 Schematic/Circuit Design

The schematic was designed using WokWi [10] to simulate the wiring of all components before hardware implementation. This simulation is used to provide a clear picture of where the pinning and wiring needed to be done on the microcontroller. Furthermore, this software is free and does not require any other additional libraries to be installed with it. Based on the figure below, the ESP32 connects via I2C to the OLED and communicates with the MAX98357A module via I2S and speaker. The two button keyboard and microphone is connected via USB and 3.5mm headphone jack interfaces respectively to the laptop. The ESP32 is powered by a 5V micro-USB cable wherelse the OLED screen is powered by the internal 3V pin located in the system. On the other hand, the speaker and audio

module requires a constant 5V power supply which can also be powered by the ESP32.

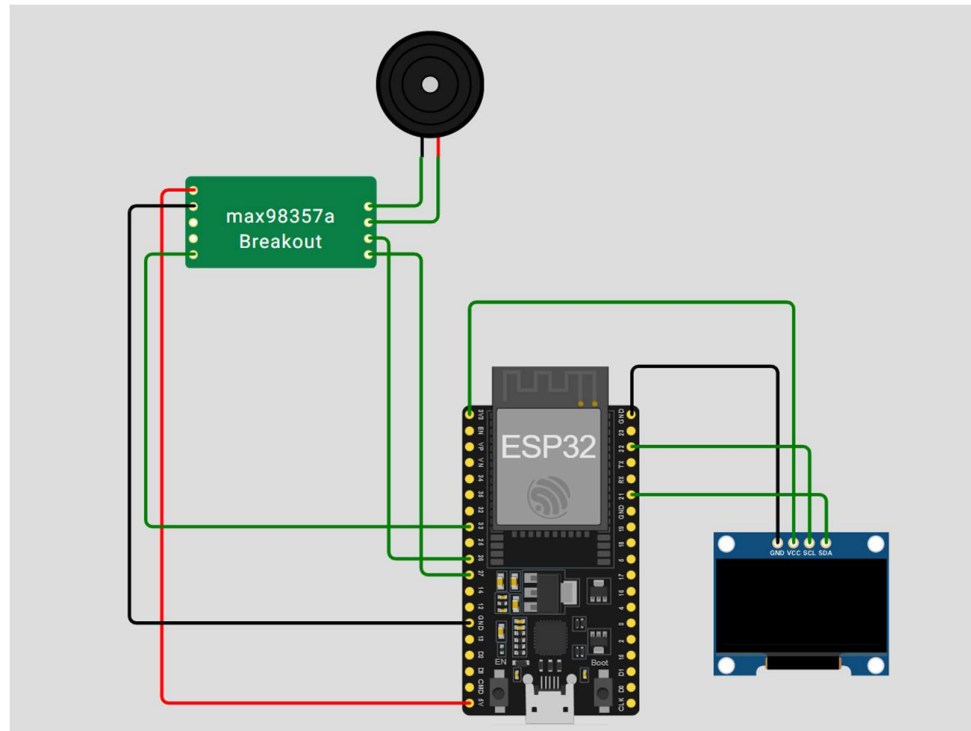


Figure 2-8 Schematic diagram of the system

Component	ESP32 Pin	Component Pin	Notes
OLED Display (SSD1306)	GPIO 21	SDA	I2C Data line
	GPIO 22	SCL	I2C Clock line
	3.3V	VCC	Power (ensure 3.3V MAX)
	GND	GND	Ground
MAX98357A (Speaker)	GPIO 33	DIN (Data In)	I2S data line
	GPIO 27	BCLK	I2S bit clock
	GPIO 26	LRC (WS/LRCK)	I2S left-right clock (word select)
	5V	VCC	Power (ensure 5V MAX)
	GND	GND	Ground

Table 2-1 Wiring Guide

2.2.3 Block Diagram

The system is represented by the block diagram below, showing the signal flow from user input to output and data logging.

Explanation:

- The student presses the two button keyboard button to begin.
- The microphone captures the spoken input.
- The ESP32 sends the data to PC via micro-usb cable to use a cloud-based speech-to-text service (OpenAI Whisper model & Lingua) to process the audio.
- Feedback is displayed on the OLED and played through the speaker.
- Usage data is logged to Microsoft Excel sheets for teacher access.

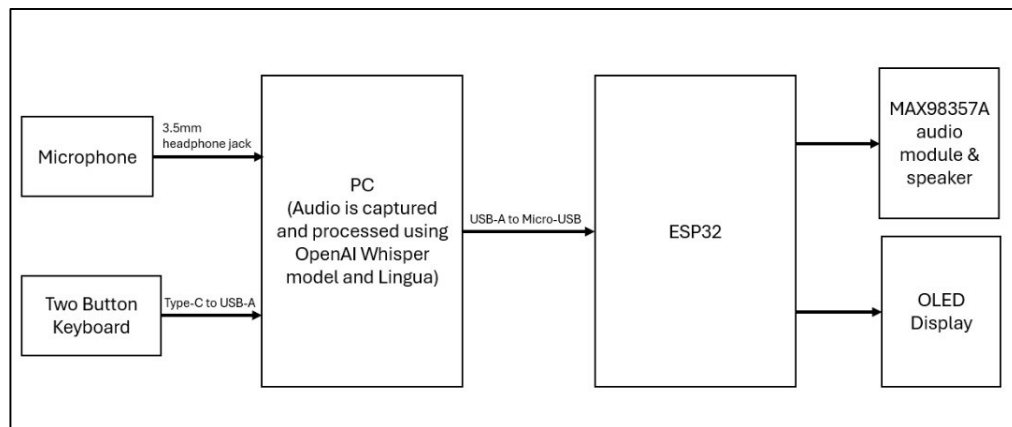


Figure 2-9 Block Diagram (ESP32)

2.2.4 Flowcharts

The process flow for students, teachers, and admin is outlined in four main flowcharts (student, teacher, admin and software mode). Each branch guides the interaction between the user and the system components. A simplified flowchart (Figure 5-1) of the overall system can be found in the Appendix.

I. Teacher Flowchart

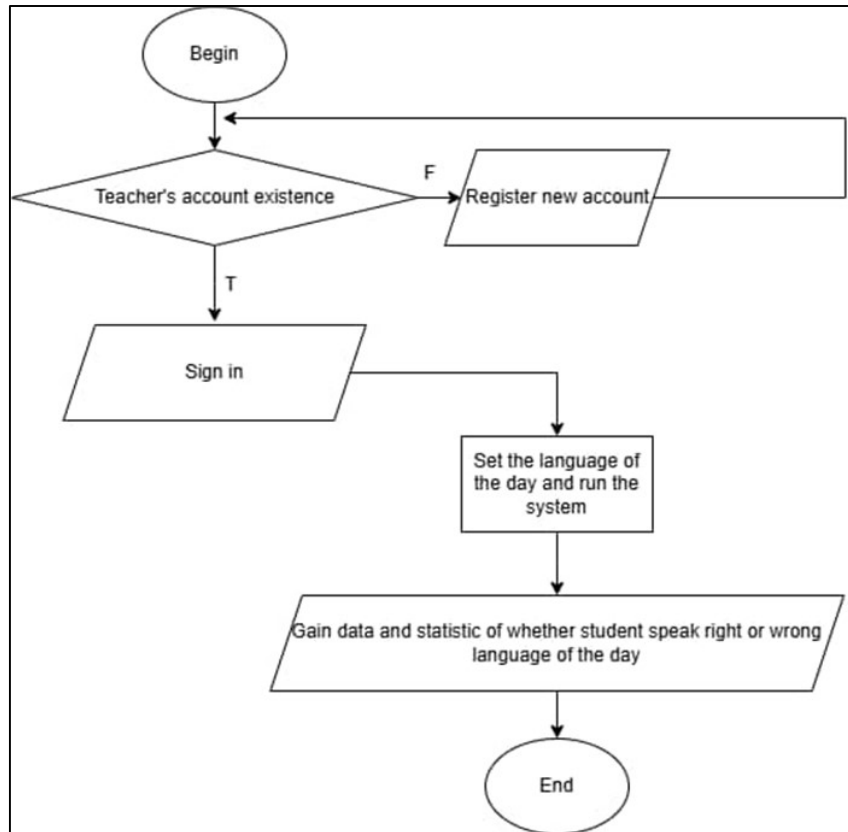


Figure 2-10 Teacher Flowchart

This flowchart (Figure 2-10) shows the process of a teacher that uses a language monitoring system for primary school students. The process begins by checking whether the teacher already has an account. If the account does not exist (False), the teacher must register a new account. If the account exists (True), the teacher proceeds to sign in. After signing in, the teacher sets the "language of the day" and activates the system. The system then runs and collects data and statistics on whether students are speaking the correct or incorrect language of the day. The process concludes after gathering this information.

II. Student Flowchart

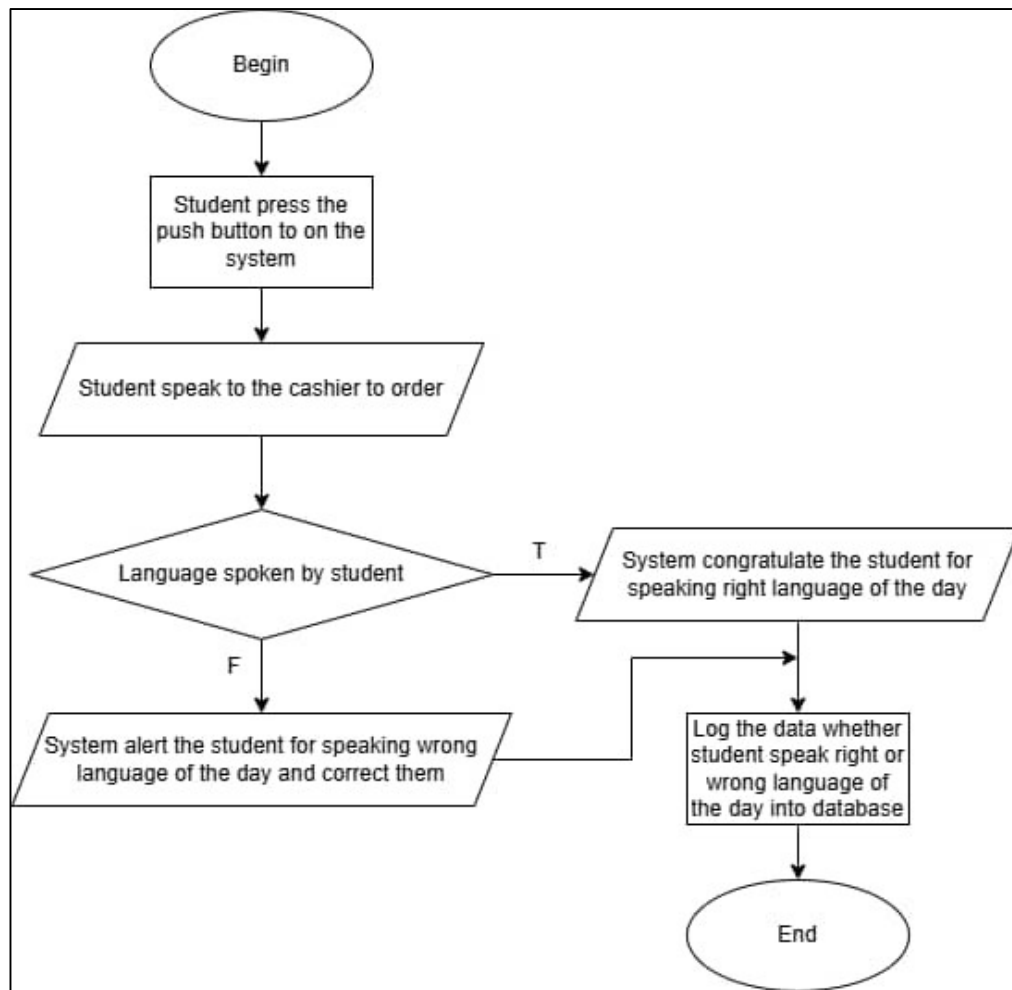


Figure 2-11 Student Flowchart

This Figure 2-11 above shows the flowchart of a student interaction system designed to encourage speaking in the assigned "language of the day." The process begins when a student presses a push button to activate the system and then speak to the cashier to place an order. The system evaluates the language spoken by the student and checks if it matches the designated language for that day. If the student uses the correct language, the system congratulates them. If the language is incorrect, the system alerts the student and provides the correct language. Regardless of the outcome, the system logs the result whether the student spoke the correct or incorrect language into a database.

III. Admin flowchart

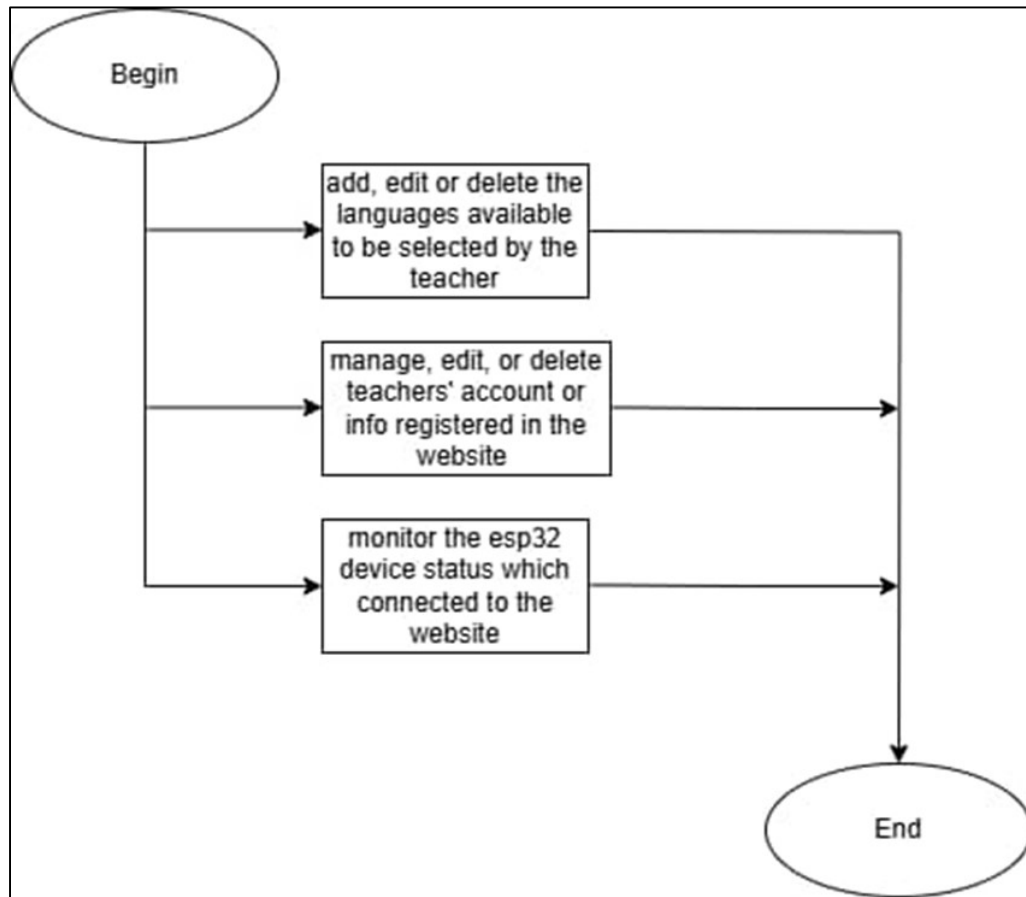


Figure 2-12 Admin Flowchart

Next, this admin flowchart (Figure 2-12) shows the functionalities of the admin of the web-based system that supports teacher-related operations and ESP32 device monitoring. It begins with a general start point and presents three parallel tasks that can be performed. Firstly, administrators can add, edit, or delete the list of languages available for teachers to select. Secondly, they can manage teacher accounts by editing or removing their information stored on the website. Thirdly, the system allows monitoring the status of ESP32 devices that are connected to the website. These tasks are handled independently and simultaneously, leading directly to the end of the process.

IV. Software Flowchart

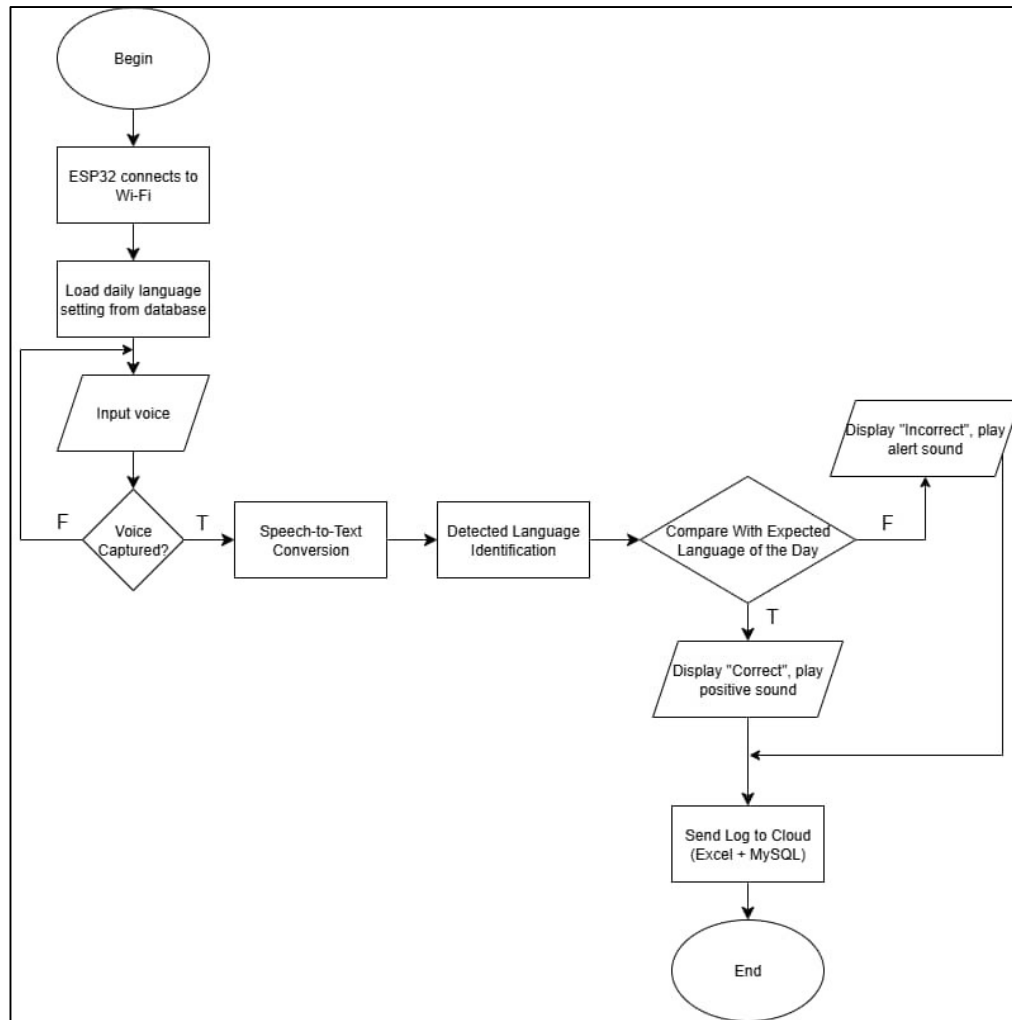


Figure 2-13 Software Flowchart

Last but not least, this flowchart (Figure 2-13) explains how an ESP32 speech-based language monitoring system works. It starts with the ESP32 connecting to Wi-Fi, then retrieving the day's language setting from the cloud database. It waits for voice input from the user. When voice is entered, it is translated from speech-to-text and language is detected. The detected language is matched against the expected language of the day. If it does, the system displays a "Correct" message with a positive sound; otherwise, it shows "Incorrect" with an alert sound. Either way, the result is logged to the cloud (via Excel and MySQL), closing the loop.

2.2.5 Software Implementation

This project includes software development alongside the hardware components. The software section plays a crucial role in ensuring the system functions smoothly, provides fast feedback to students, and allows both teachers and administrators to monitor language usage. The main programming was done using Arduino IDE for the ESP32 microcontroller whereas the Python script in PC handles input from the microphone, processes the speech, verifies the language used using Whisper voice recognition model, and gives feedback through the OLED display and speaker via the ESP32. To detect speech, the system applies OpenAI Whisper voice recognition model and Lingua compares the detected language with the expected language of the day. If the language is correct, a positive message appears on the OLED screen and a confirmation sound is played. If incorrect, a different message is shown. All results are recorded in the system's database. Teachers can later export these records into Microsoft Excel format for analysis and tracking purposes. A web interface was developed using PHP and MySQL to manage the system. Teachers can log in, set the daily language, and view usage logs. Administrators can monitor system activity and manage basic system settings.

All voice activity is saved in a structured **MySQL database**. It stores:

- Detected language and time
- Whether it was correct or not
- Daily language setting by teacher
- System errors (e.g. mic not working)

This structure makes the system more reliable and allows easy tracking of both performance and technical issues. In Figure 2-14, displays Composer [11], the dependency management tool for PHP. Composer is essential in this project for managing and installing various PHP libraries, including PHPMailer, ensuring that all necessary components are available and properly loaded for the web interface functionality.

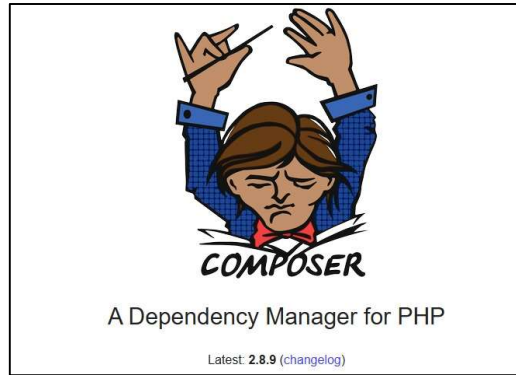


Figure 2-14 Composer Tool for PHP

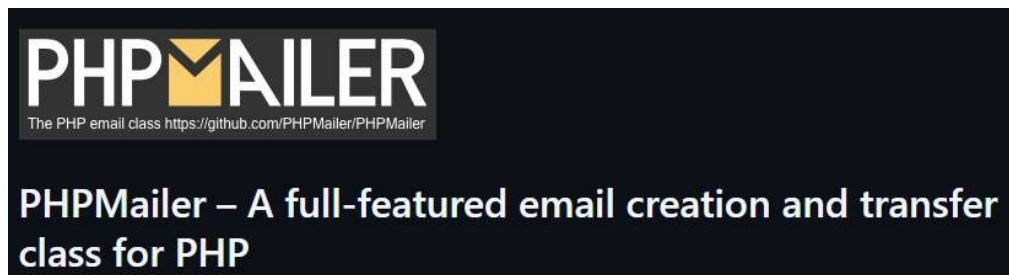


Figure 2-15 PHPMailer

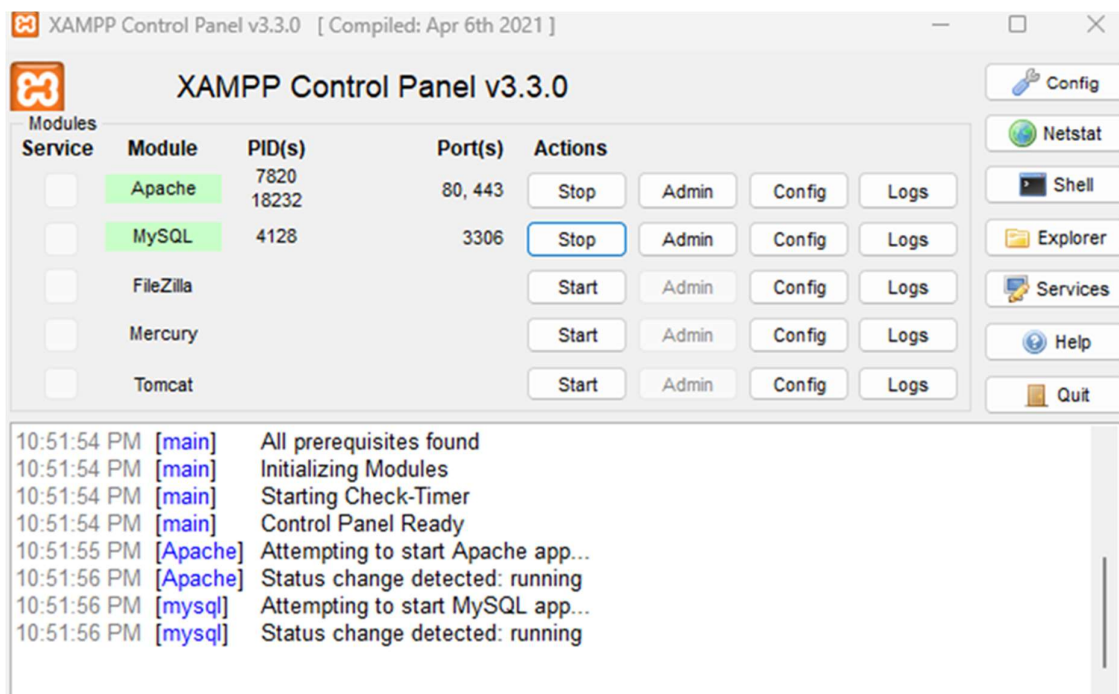


Figure 2-16 XAMPP Control Panel

In Figure 2-15, PHPMailer [12] is a widely-used PHP class designed for sending emails. This project utilizes PHPMailer to handle the reliable and secure delivery of automated emails, such as password reset links to users. As in Figure 2-16, XAMPP Control Panel [13] serves as the local server environment for the web-based system. It is used to manage and run essential services like Apache (web server) and MySQL (database server), allowing for local development and testing of the PHP and database functionalities.



Size	Parameters	English-only model	Multilingual model	Required VRAM	Relative speed
tiny	39 M	<code>tiny.en</code>	<code>tiny</code>	~1 GB	~10x
base	74 M	<code>base.en</code>	<code>base</code>	~1 GB	~7x
small	244 M	<code>small.en</code>	<code>small</code>	~2 GB	~4x
medium	769 M	<code>medium.en</code>	<code>medium</code>	~5 GB	~2x
large	1550 M	N/A	<code>large</code>	~10 GB	1x
turbo	809 M	N/A	<code>turbo</code>	~6 GB	~8x

Figure 2-17 OpenAI Whisper model configuration

Based on the figure above, the configuration section for the OpenAI Whisper model [14] within the Python script (`language_monitor.py`). This open-source speech recognition model is used for transcribing spoken audio into text and automatically detecting the language. The figure highlights its integration, showing options for different model sizes (e.g., 'tiny', 'base'), each with varying parameters, required VRAM, and relative speed, which allows for optimization based on performance needs.

2.2.6 Software System Screenshots

I. Admin Section

- a) Admin Login Page (This page allows admin users to access the control system securely.)

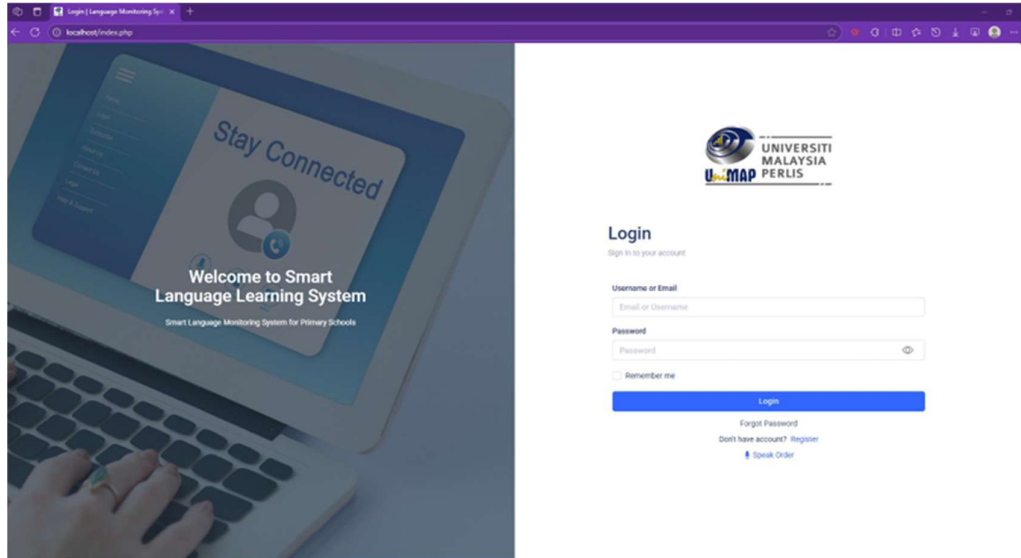


Figure 2-18 Admin Login Page

- b) Admin Dashboard (Overview of system users and device status)

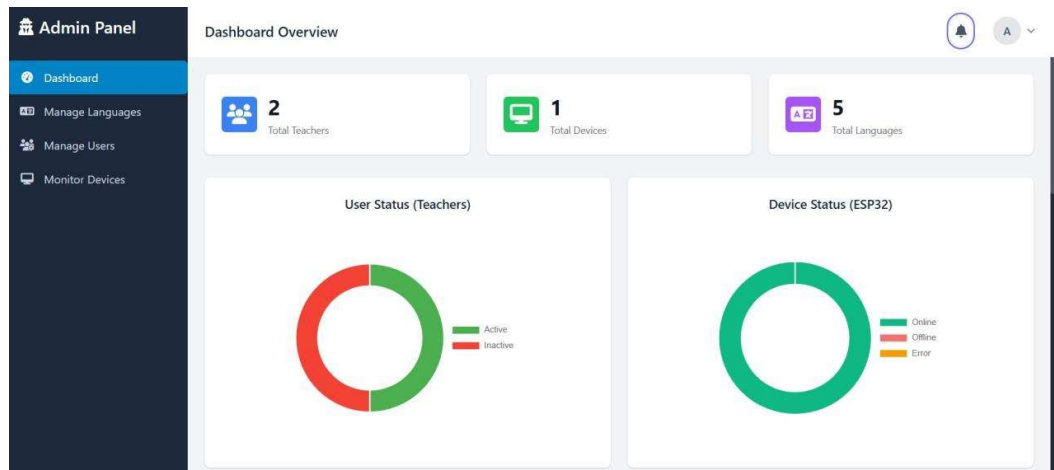


Figure 2-19 Admin Dashboard

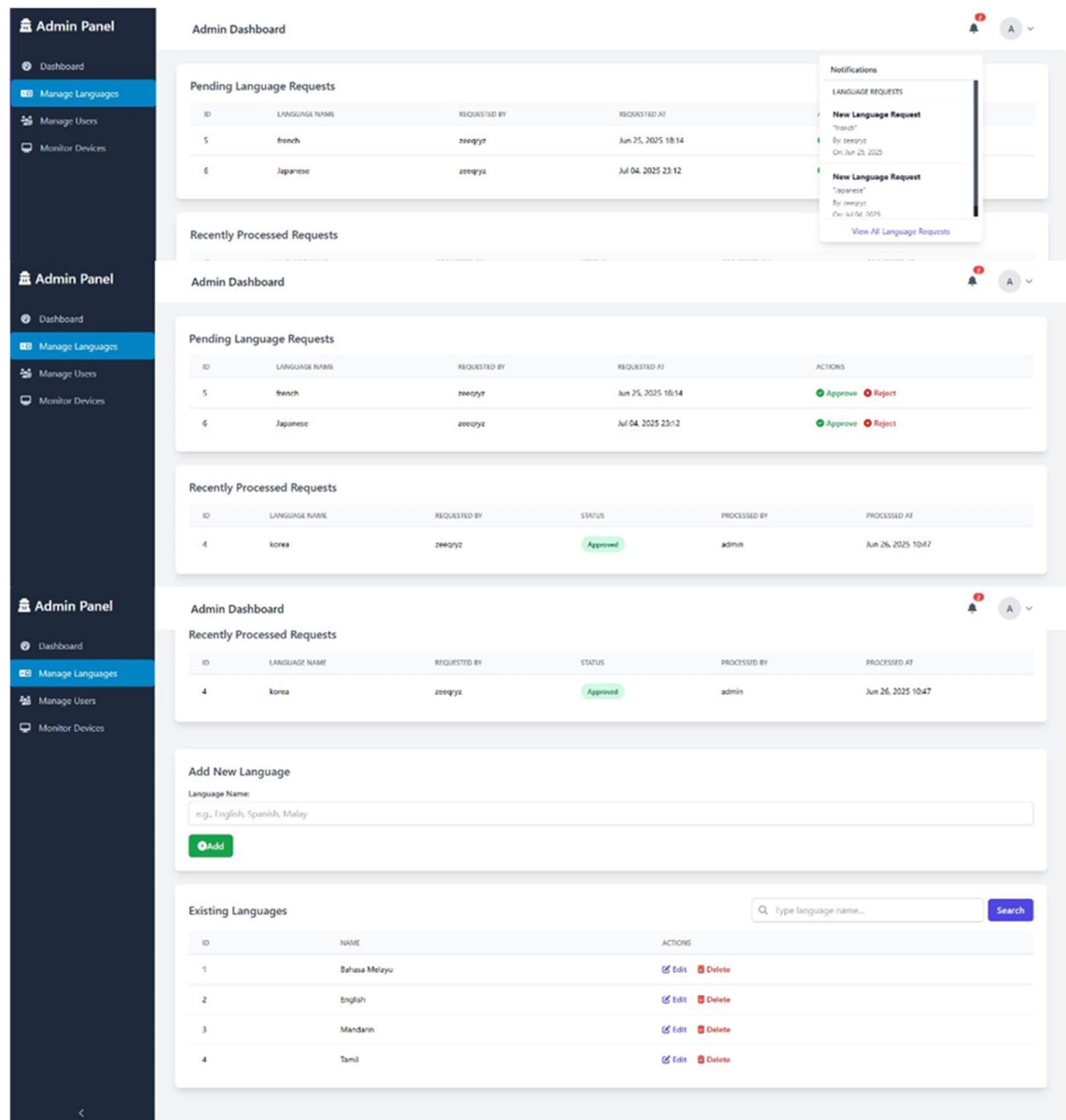


Figure 2-20 Admin Privileges

Based on the figure above, it shows the administrative interface for managing languages within the system. This page allows administrators to add new languages, view existing languages with their details, and perform actions such as approving or rejecting new language requests, as well as editing or deleting existing language entries.

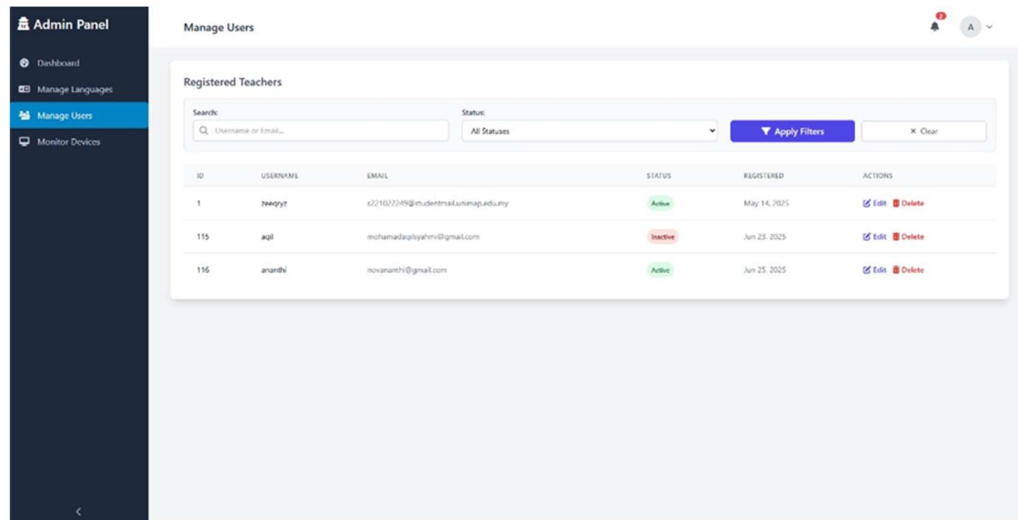


Figure 2-21 Manage Users Interface

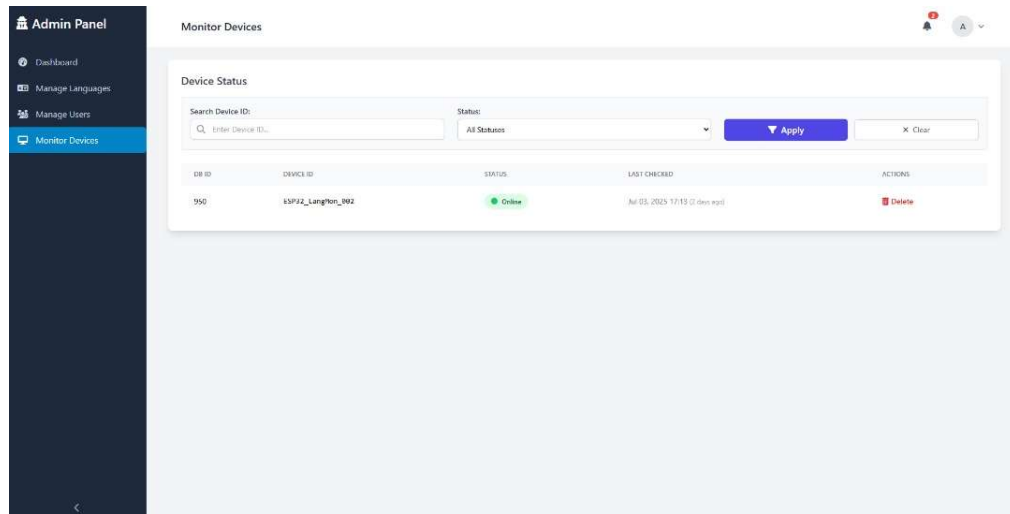


Figure 2-22 Monitor Device Interface

As shown in Figure 2-21, the 'Manage Users' interface is within the Admin Panel where it provides administrators with tools to view all registered teachers, including their username, email, and current status (active/inactive). It also allows filtering by status and provides options to edit teacher profiles or delete user accounts. The following figure displays the 'Monitor Devices' interface within the Admin Panel. This page provides administrators with a real-time overview of the status of connected ESP32 devices, showing their unique ID, current operational status (e.g., online), and the last time their status was updated. It also includes search and filter functionalities to manage device monitoring efficiently.

c) Statistics of teacher registrations and language additions by date.

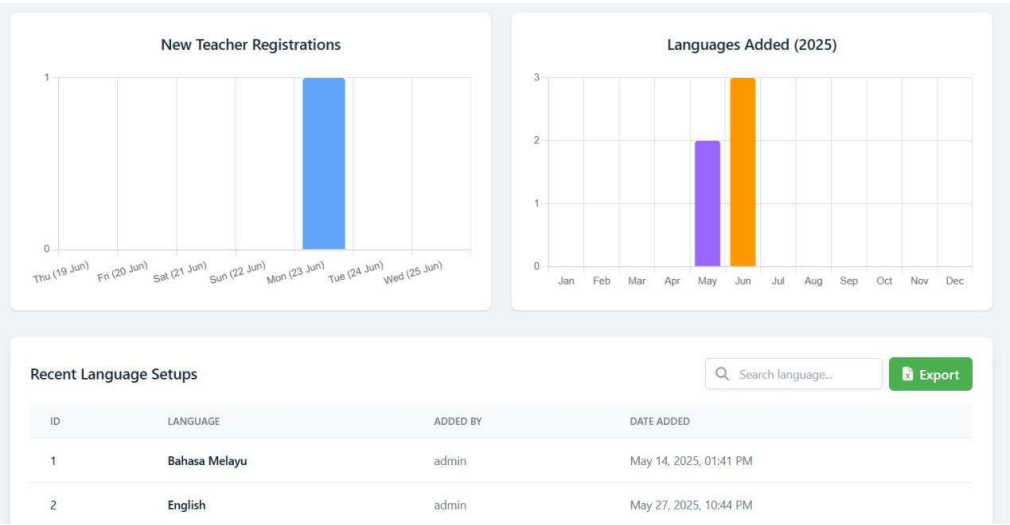


Figure 2-23 Teacher registrations statistics

d) Logs of password reset requests and teacher-set daily language history (with export options).

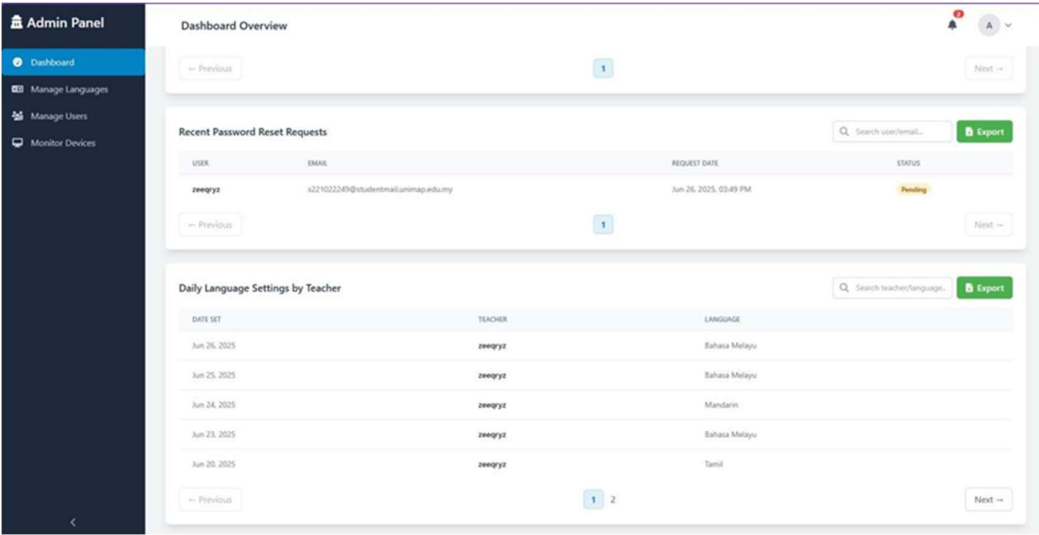
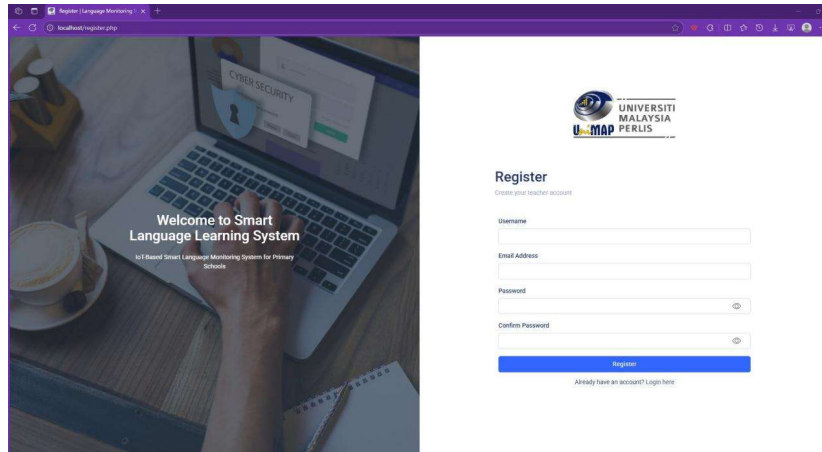


Figure 2-24 Logs of password

II. Teacher Section

- a) Teacher Registration / Login (Teachers are required to register or log in to set daily language and view results)



Register

Create your teacher account

Username

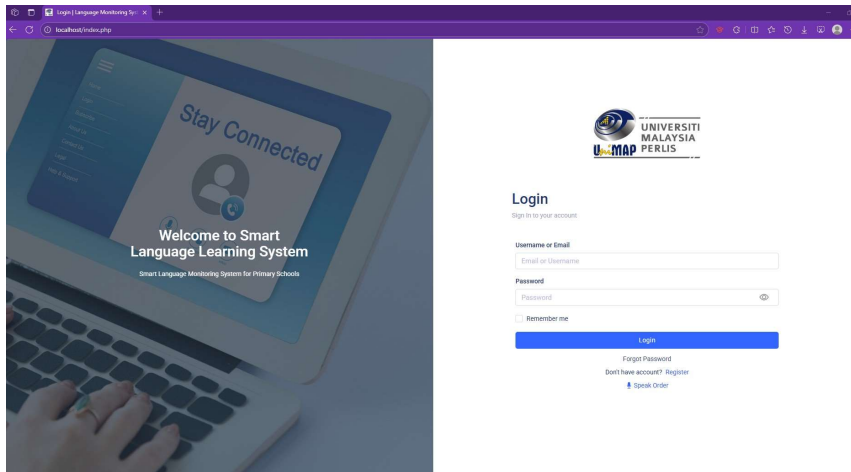
Email Address

Password

Confirm Password

Already have an account? [Login here](#)

Figure 2-25 Teacher Registration



Login

Sign in to your account

Username or Email

Password

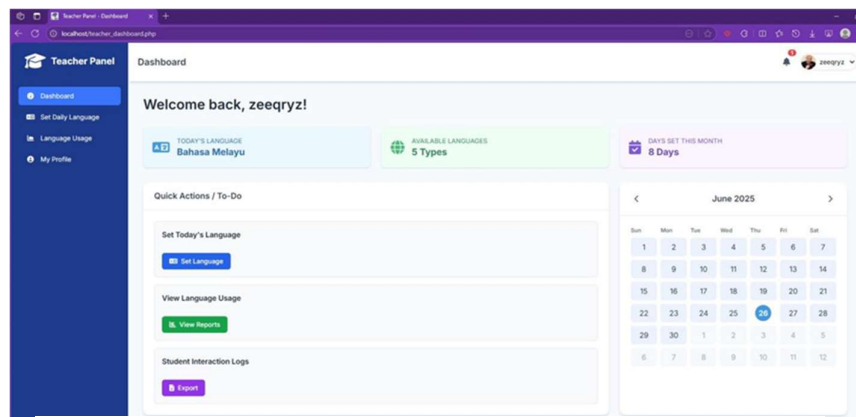
☐ Remember me

Forgot Password? [Forgot Password](#)

Don't have an account? [Register](#)

[Speak Order](#)

Figure 2-26 Teacher Login



Teacher Panel

Dashboard

Welcome back, zeeqryz!

TODAY'S LANGUAGE
Bahasa Malaysia

AVAILABLE LANGUAGES
5 Types

DAYS SET THIS MONTH
8 Days

Quick Actions / To-Do

Set Today's Language

View Language Usage

Student Interaction Logs

June 2025

Sun	Mon	Tue	Wed	Thu	Fri	Sat
1	2	3	4	5	6	7
8	9	10	11	12	13	14
15	16	17	18	19	20	21
22	23	24	25	26	27	28
29	30	1	2	3	4	5
6	7	8	9	10	11	12

Figure 2-27 Teacher Dashboard (First front webpage overview)

- b) Set Daily Language (Teachers can select a date and choose one of the available languages from a dropdown list to update the system's daily language setting.)

Teacher Panel Set Daily Language

Set Language of the Day

Select Date: 26/06/2025

Select Language for 2025-06-26: Bahasa Melayu

Select

Important Notice:

- Use this form to set the language for a specific day.
- For any given day, only ONE language can be set per day. If you set a language for a date that already has one, it will be updated to your new selection.
- Use the "Actions" column in the history table below to directly edit or delete an existing entry.

Daily Language Settings History

DATE	LANGUAGE	SET BY TEACHER	ACTIONS
2025-07-04	English	zeeqyz	Edit Delete
2025-07-03	Mandarin	zeeqyz	Edit Delete
2025-07-02	Bahasa Melayu	zeeqyz	Edit Delete
2025-07-01	Tamil	zeeqyz	Edit Delete
2025-06-27	English	zeeqyz	Edit Delete
2025-06-26	Tamil	zeeqyz	Edit Delete
2025-06-25	Bahasa Melayu	zeeqyz	Edit Delete
2025-06-24	Mandarin	zeeqyz	Edit Delete
2025-06-23	Bahasa Melayu	zeeqyz	Edit Delete
2025-06-20	Tamil	zeeqyz	Edit Delete

Showing 1 to 10 of 15 entries

Figure 2-28 Teacher set daily language

Teacher Panel My Profile

My Profile

Change Photo

Username: zeeqyz

Email: s22f02224@studentmail.unimap.edu.my

Change Password (Optional)

Current Password

New Password (Min 8 characters)

Confirm New Password

Cancel Save Changes

Figure 2-29 Teacher Profile Page

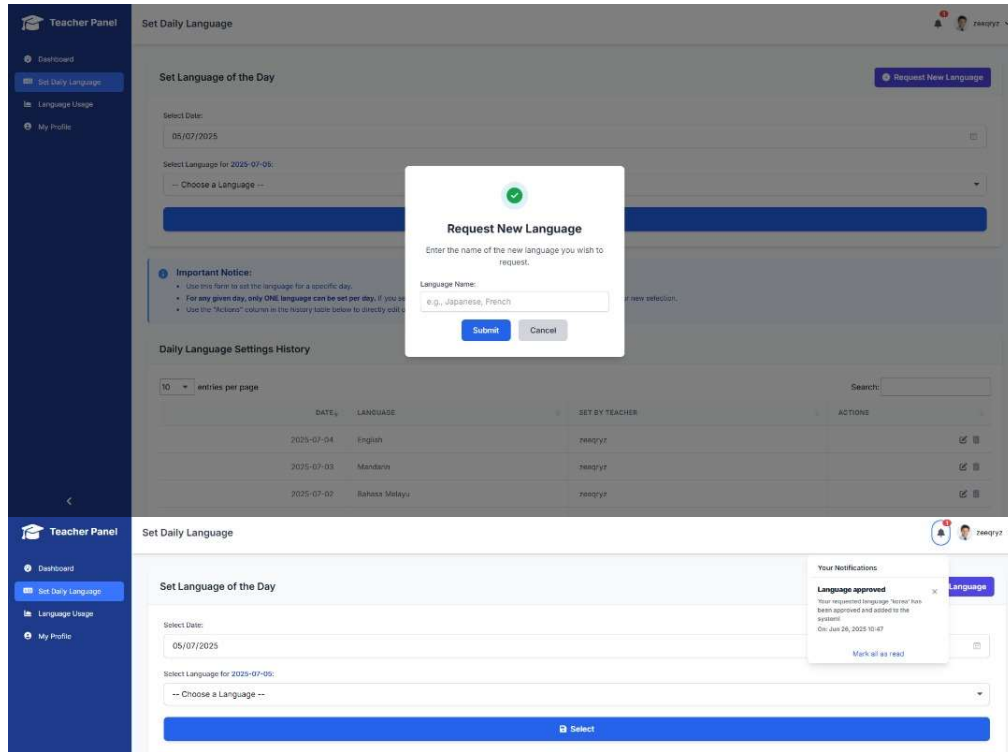


Figure 2-30 Request New Language

In Figure 2-29, it displays the 'My Profile' page within the Teacher Panel. This interface allows teachers to update their personal information, including their username and email, and to change their profile picture. It also provides fields for updating their password, requiring the current password for security. Where else in Figure 2-30, it illustrates the 'Request New Language' modal, accessible from the 'Set Daily Language' page. This pop-up allows teachers to submit a request for a new language to be added to the system by entering its name, which then requires administrative approval.

- c) Password Settings (Configure passwords when password is forgotten or account issues)

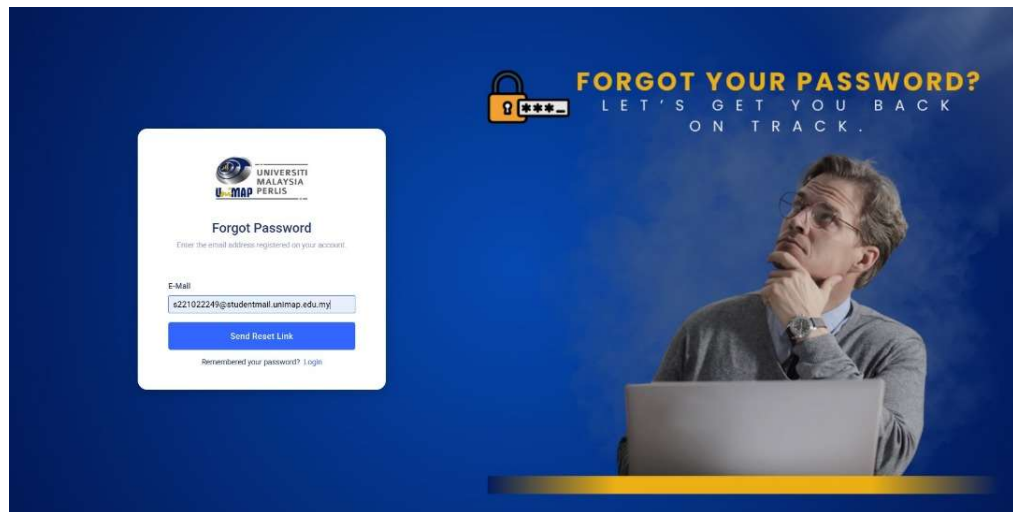


Figure 2-31 Forgot Password Interface

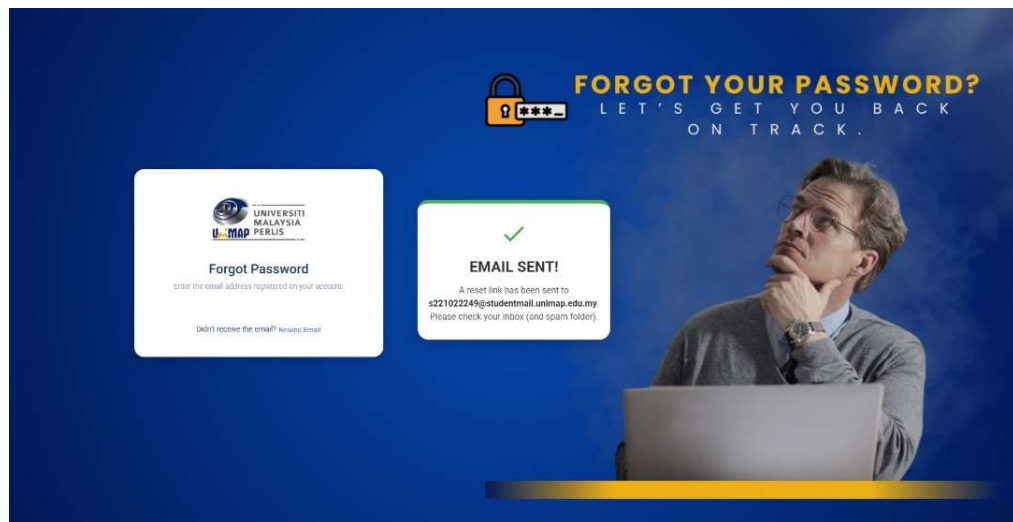


Figure 2-32 Password Confirmation

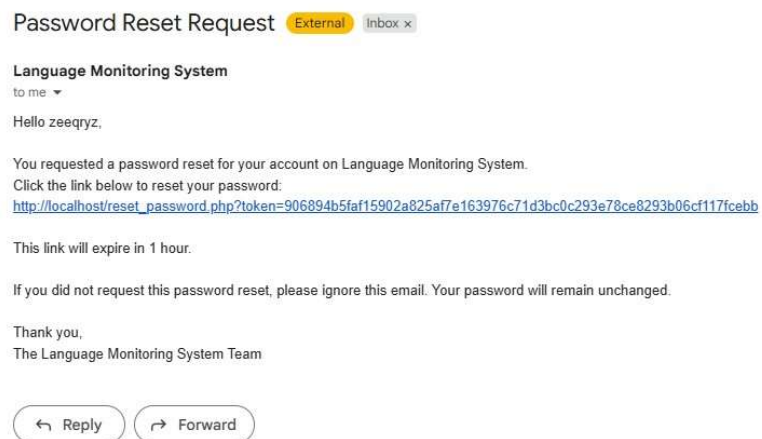


Figure 2-33 Email Screenshot

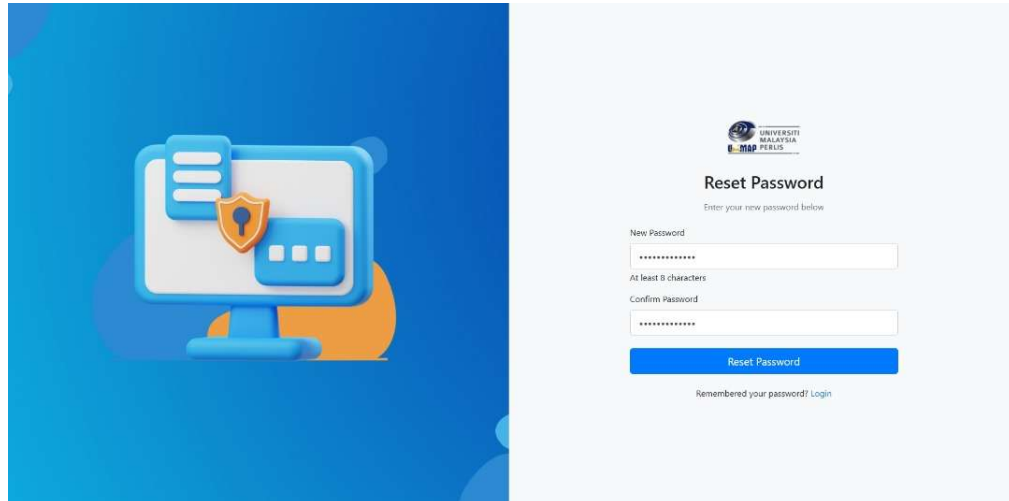
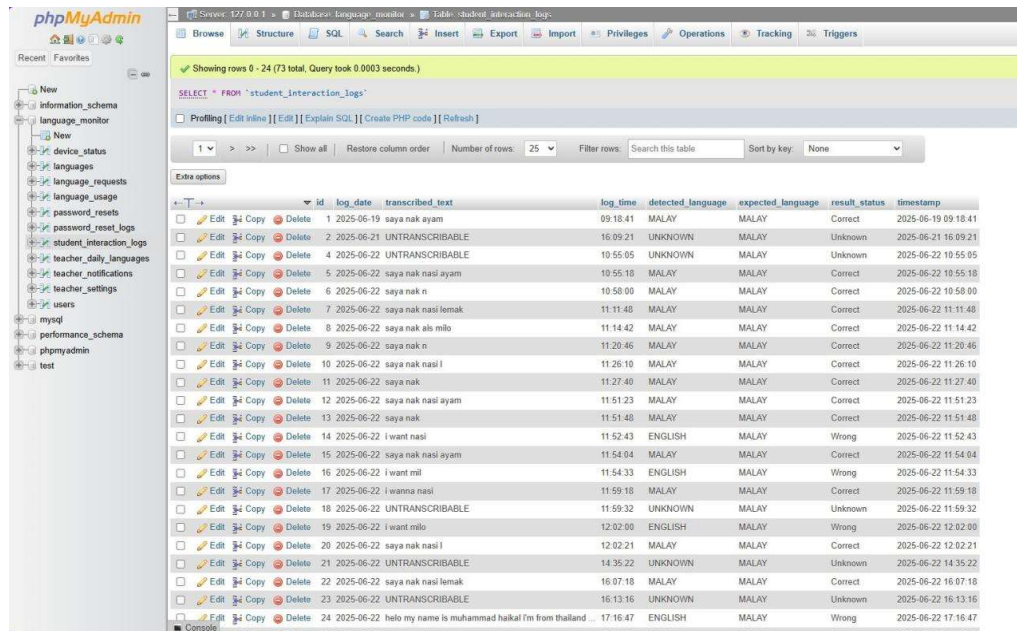


Figure 2-34 Reset password page

The figures above show the process of what to do when the password is forgotten or needs to be changed. In Figure 2-31, it illustrates the 'Forgot Password' interface where users can initiate the password reset process by entering their registered email address. Then, a confirmation screen will be displayed to the user after a password reset request has been successfully sent to their registered email address as shown in Figure 2-32. Next, Figure 2-33 shows the screenshot of the automated email received by the user, containing the secure link to reset their password for the Language Monitoring System. Lastly, users can enter and confirm their new password after clicking the reset link received via email in the 'Reset Password' page such as in Figure 2-34.

III. Data Logging Section

- a) Database Structure (After student audio is captured and is processed in the OpenAI Whisper voice recognition model, all the data is stored into a database which includes speech recognition results, including timestamp, detected & expected language and status.)



id	log_date	transcribed_text	log_time	detected_language	expected_language	result_status	timestamp
1	2025-06-19	saya nak ayam	09:18:41	MALAY	MALAY	Correct	2025-06-19 09:18:41
2	2025-06-21	UNTRANSCRIBABLE	16:09:21	UNKNOWN	MALAY	Unknown	2025-06-21 16:09:21
4	2025-06-22	UNTRANSCRIBABLE	10:55:05	UNKNOWN	MALAY	Unknown	2025-06-22 10:55:05
5	2025-06-22	saya nak nasi ayam	10:55:18	MALAY	MALAY	Correct	2025-06-22 10:55:18
6	2025-06-22	saya nak n	10:58:00	MALAY	MALAY	Correct	2025-06-22 10:58:00
7	2025-06-22	saya nak nasi lemak	11:11:48	MALAY	MALAY	Correct	2025-06-22 11:11:48
8	2025-06-22	saya nak als milo	11:14:42	MALAY	MALAY	Correct	2025-06-22 11:14:42
9	2025-06-22	saya nak n	11:20:46	MALAY	MALAY	Correct	2025-06-22 11:20:46
10	2025-06-22	saya nak nasi i	11:26:10	MALAY	MALAY	Correct	2025-06-22 11:26:10
11	2025-06-22	saya nak	11:27:40	MALAY	MALAY	Correct	2025-06-22 11:27:40
12	2025-06-22	saya nak nasi ayam	11:51:23	MALAY	MALAY	Correct	2025-06-22 11:51:23
13	2025-06-22	saya nak	11:51:48	MALAY	MALAY	Correct	2025-06-22 11:51:48
14	2025-06-22	i want nasi	11:52:43	ENGLISH	MALAY	Wrong	2025-06-22 11:52:43
15	2025-06-22	saya nak nasi ayam	11:54:04	MALAY	MALAY	Correct	2025-06-22 11:54:04
16	2025-06-22	i want mil	11:54:33	ENGLISH	MALAY	Wrong	2025-06-22 11:54:33
17	2025-06-22	i wanna nasi	11:59:18	MALAY	MALAY	Correct	2025-06-22 11:59:18
18	2025-06-22	UNTRANSCRIBABLE	11:59:32	UNKNOWN	MALAY	Unknown	2025-06-22 11:59:32
19	2025-06-22	i want milo	12:02:00	ENGLISH	MALAY	Wrong	2025-06-22 12:02:00
20	2025-06-22	saya nak nasi i	12:02:21	MALAY	MALAY	Correct	2025-06-22 12:02:21
21	2025-06-22	UNTRANSCRIBABLE	14:35:22	UNKNOWN	MALAY	Unknown	2025-06-22 14:35:22
22	2025-06-22	saya nak nasi lemak	16:07:18	MALAY	MALAY	Correct	2025-06-22 16:07:18
23	2025-06-22	UNTRANSCRIBABLE	16:13:16	UNKNOWN	MALAY	Unknown	2025-06-22 16:13:16
24	2025-06-22	hello my name is muhammad haikal i'm from thailand ...	17:16:47	ENGLISH	MALAY	Wrong	2025-06-22 17:16:47

Figure 2-35 Logs data in database

- b) Backend Database Structure (Shows the main tables used for logging interactions, managing users, language settings, and device status.)

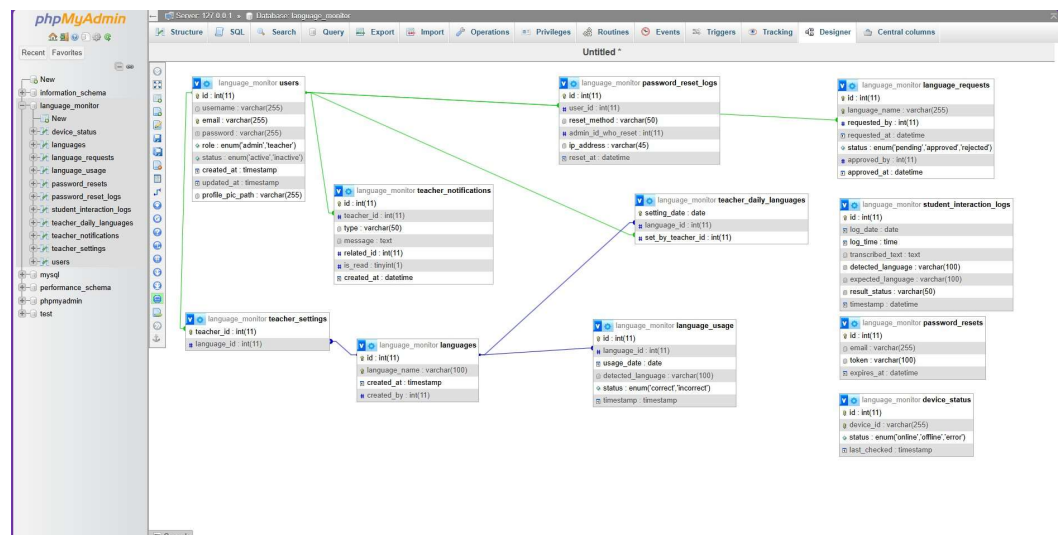


Figure 2-36 Database Structure – Core Tables and Relationships

c) Student Voice Logs (Can be exported by teachers in Microsoft Excel format for easier viewing and record-keeping)

	A	B	C	D	E	F	G	H
13	27/6/2025	23:11:52 berapakah harga nasi ayam	MALAY MALAY Correct	27/6/2025 23:11				
14	27/6/2025	23:11:26 UNTRANSCRIBABLE	UNKNOWN MALAY Unknown	27/6/2025 23:11				
15	27/6/2025	23:09:53 ni air ni berapa kupang	MALAY ENGLISH Wrong	27/6/2025 23:09				
16	27/6/2025	23:09:30 how much does chicken rice cost	ENGLISH ENGLISH Correct	27/6/2025 23:09				
17	27/6/2025	23:07:42 berapakah harga nasi ayam	MALAY ENGLISH Wrong	27/6/2025 23:07				
18	27/6/2025	22:59:35 how much is the chicken rice that I order	ENGLISH ENGLISH Correct	27/6/2025 22:59				
19	27/6/2025	22:58:46 how much is the chicken rice that I order	ENGLISH ENGLISH Correct	27/6/2025 22:58				
20	27/6/2025	22:58:15 UNTRANSCRIBABLE	UNKNOWN ENGLISH Unknown	27/6/2025 22:58				
21	27/6/2025	22:47:53 how much does chicken rice	ENGLISH ENGLISH Correct	27/6/2025 22:47				
22	27/6/2025	22:46:44 hi	ENGLISH ENGLISH Correct	27/6/2025 22:46				
23	27/6/2025	22:45:01 NO_SPEECH_DETECTED	N/A ENGLISH NoSpeech	27/6/2025 22:45				
24	27/6/2025	22:44:38 MIC_ERROR: 'NoneType' object has no attribute 'close'	N/A ENGLISH MicError	27/6/2025 22:44				
25	27/6/2025	22:34:50 MIC_ERROR: 'NoneType' object has no attribute 'close'	N/A ENGLISH MicError	27/6/2025 22:34				
26	27/6/2025	22:34:21 MIC_ERROR: 'NoneType' object has no attribute 'close'	N/A ENGLISH MicError	27/6/2025 22:34				
27	27/6/2025	22:33:47 MIC_ERROR: 'NoneType' object has no attribute 'close'	N/A ENGLISH MicError	27/6/2025 22:33				
28	26/6/2025	23:58:00 NO_SPEECH_DETECTED	N/A TAMIL NoSpeech	26/6/2025 23:58				
29	26/6/2025	23:50:34 berapakah harga nasi ayam yang saya pesan	MALAY TAMIL Wrong	26/6/2025 23:50				
30	26/6/2025	23:50:13 აბა ბარძიშ ღარიანი ხაჭაპური რისკითაა?	TAMIL TAMIL Correct	26/6/2025 23:50				
31	26/6/2025	23:49:10 berapakah harga nasi ayam yang saya pesan	MALAY CHINESE Wrong	26/6/2025 23:49				
32	26/6/2025	23:48:31 what is the chief and push	ENGLISH CHINESE Wrong	26/6/2025 23:48				
33	26/6/2025	23:34:45 Barack Obama says Susan	ENGLISH CHINESE Wrong	26/6/2025 23:34				
34	26/6/2025	23:34:00 me "c,"y_e,jek-kaŋa"e'a	CHINESE CHINESE Correct	26/6/2025 23:34				
35	26/6/2025	23:32:35 who's your cheating	ENGLISH CHINESE Wrong	26/6/2025 23:32				
36	26/6/2025	23:31:39 saya nak makan roti canal	MALAY TAMIL Wrong	26/6/2025 23:31				
37	26/6/2025	23:30:49 აზღაგანო დასახლებაში არასრულწლოვანების განთავსება შეემატა	TAMIL TAMIL Correct	26/6/2025 23:30				
38	26/6/2025	23:30:04 unike sosial	MALAY TAMIL Wrong	26/6/2025 23:30				
39	26/6/2025	23:28:57 saya nak makan sushi	MALAY ENGLISH Wrong	26/6/2025 23:28				
40	26/6/2025	23:27:33 i want to eat sushi near the pronunciation English US delete up	ENGLISH ENGLISH Correct	26/6/2025 23:27				
41	26/6/2025	23:26:59 i want to wait sushi	ENGLISH ENGLISH Correct	26/6/2025 23:26				
42	26/6/2025	23:24:40 saya nak makan sushi	MALAY MALAY Correct	26/6/2025 23:24				
43	26/6/2025	23:24:20 UNTRANSCRIBABLE	UNKNOWN MALAY Unknown	26/6/2025 23:24				
44	26/6/2025	23:23:43 i want to eat sushi	ENGLISH MALAY Wrong	26/6/2025 23:23				
45	26/6/2025	23:13:42 saya nak makan nasi ayam bersama minuman limau	MALAY MALAY Correct	26/6/2025 23:13				
46	26/6/2025	23:12:12 uncle please calculat	ENGLISH MALAY Wrong	26/6/2025 23:12				
47	26/6/2025	23:11:43 patchy Tulum ki recipe	ENGLISH MALAY Wrong	26/6/2025 23:11				
48	26/6/2025	23:09:53 uncle please calculat	ENGLISH ENGLISH Correct	26/6/2025 23:09				
49	26/6/2025	23:04:29 uncle please calculat	MALAY ENGLISH Wrong	26/6/2025 23:04				

29

	A	B	C	D	E
1					
2	Date	Detected Language	Status	Count	
3	1/7/2025	CHINESE	incorrect	3	
4	1/7/2025	ENGLISH	incorrect	8	
5	1/7/2025	MALAY	correct	4	
6	1/7/2025	N/A	incorrect	3	
7	1/7/2025	TAMIL	correct	2	
8	1/7/2025	TAMIL	incorrect	2	
9	1/7/2025	UNKNOWN	incorrect	28	
10	2/7/2025	CHINESE	incorrect	1	
11	2/7/2025	ENGLISH	correct	11	
12	2/7/2025	ENGLISH	incorrect	30	
13	2/7/2025	MALAY	correct	33	
14	2/7/2025	MALAY	incorrect	6	
15	2/7/2025	N/A	incorrect	4	
16	2/7/2025	TAMIL	incorrect	2	
17	2/7/2025	UNKNOWN	incorrect	21	
18	3/7/2025	CHINESE	incorrect	3	
19	3/7/2025	ENGLISH	correct	3	
20	3/7/2025	ENGLISH	incorrect	12	
21	3/7/2025	MALAY	incorrect	27	
22	3/7/2025	N/A	incorrect	8	
23	3/7/2025	TAMIL	correct	4	
24	3/7/2025	TAMIL	incorrect	1	
25	3/7/2025	UNKNOWN	incorrect	7	

Figure 2-38 Results Log

2.2.7 3-Dimensional Design

The casing was designed using Tinkercad to protect the components and make the system suitable for school use. The design includes a speaker hole, OLED screen window, and wire hole.

Design Considerations:

- Display is placed above to allow easy readability.
- Ports are properly aligned to avoid wire strain.
- The microphone is placed facing outward for clear voice detection.
- Speaker is placed on side for proper hearing of user

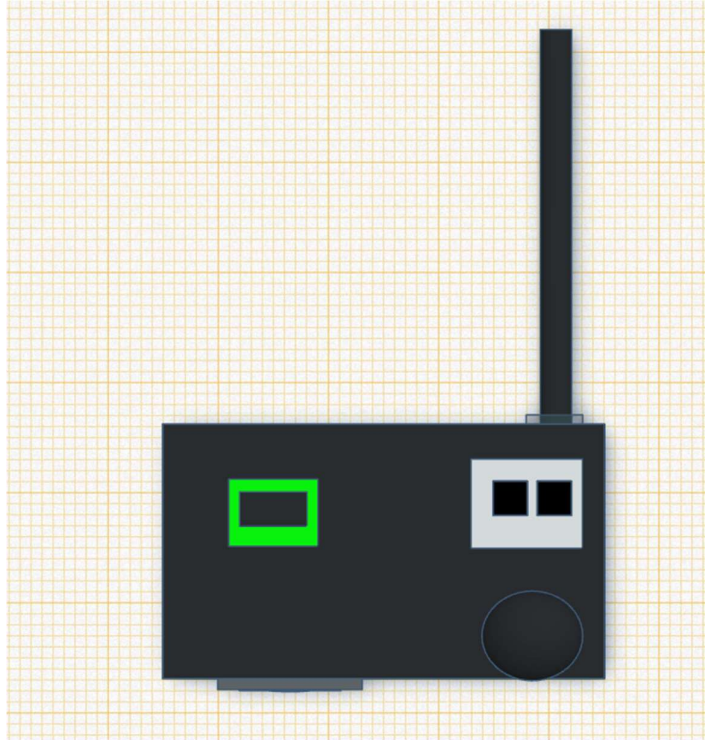


Figure 2-39 Front View of 3D Design

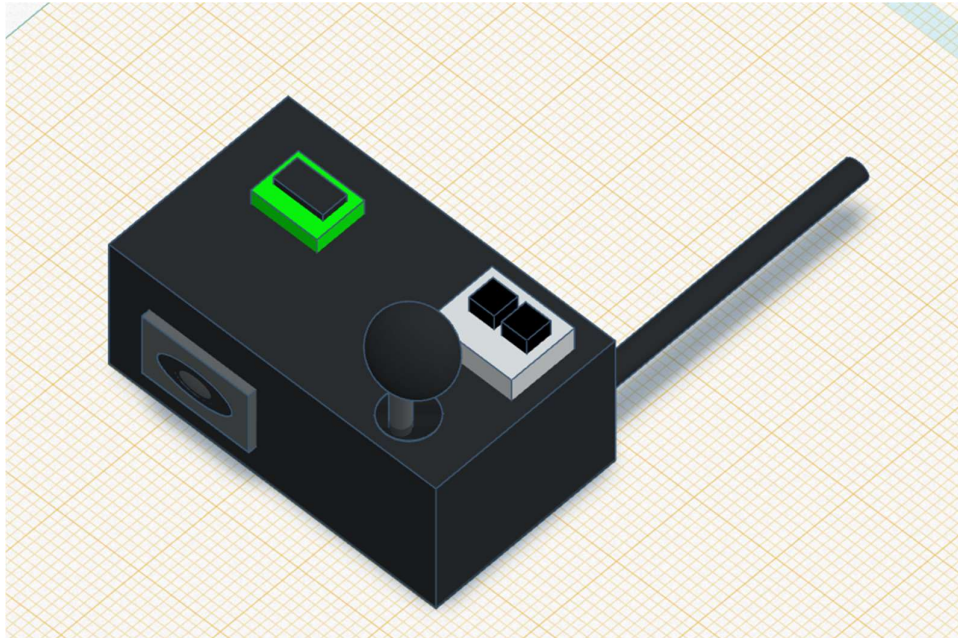


Figure 2-40 Side View of 3D Design

2.2.8 Prototype Design

After so many test runs and trials, these are the final images of our prototype system, IoT Based Smart Language Monitoring System For Primary Schools. Below are the images of the front and side of the prototype.



Figure 2-41 Front View of prototype



Figure 2-42 Side View of Prototype

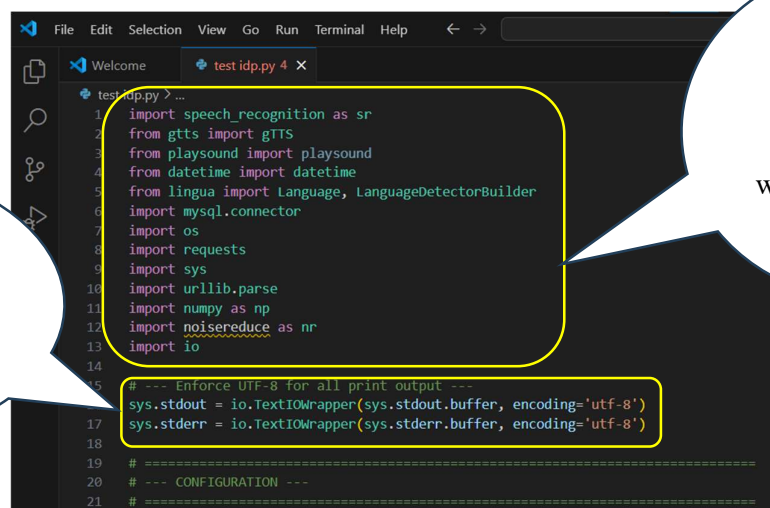
2.2.9 Coding

Coding plays a very important role in our project as it is the backbone for the whole system. Coding or in other words, programming is the process of writing instructions that a computer, motherboard or a chipset can understand and execute. These instructions can be written in many programming languages such as Python, Java, JavaScript and many more in this modern age. These instructions tell the computer what to do step by step. In the upcoming sections, an explanation will be provided for the python script and ESP32 coding needed for this system.

a) Python Source Code

The main goal of this python source code is to implement AI into this system by processing the language detected and transcribing it along with displaying it to the users. This python script can be executed using the built-in terminal provided by the Microsoft VisualCode software. The following goals for the script are as follows:

- 1) Capture the spoken audio via a microphone
- 2) Transcribe using OpenAI Whisper
- 3) Detect the language using Lingua
- 4) Compares to the target language of the day from MySQL
- 5) Sends feedback to an ESP32 device
- 6) Logs everything into a MySQL database



```
1 import speech_recognition as sr
2 from gtts import gTTS
3 from playsound import playsound
4 from datetime import datetime
5 from lingua import Language, LanguageDetectorBuilder
6 import mysql.connector
7 import os
8 import requests
9 import sys
10 import urllib.parse
11 import numpy as np
12 import noisereduce as nr
13 import io
14
15 # --- Enforce UTF-8 for all print output ---
16 sys.stdout = io.TextIOWrapper(sys.stdout.buffer, encoding='utf-8')
17 sys.stderr = io.TextIOWrapper(sys.stderr.buffer, encoding='utf-8')
18
19 # ===== CONFIGURATION =====
20 # --- CONFIGURATION ---
21 # =====
```

This is to ensure that all console output will support UTF-8 (important for multilingual output)

Libraries are needed to be imported to run the system without errors and accordingly.

Figure 2-43 Python Section A

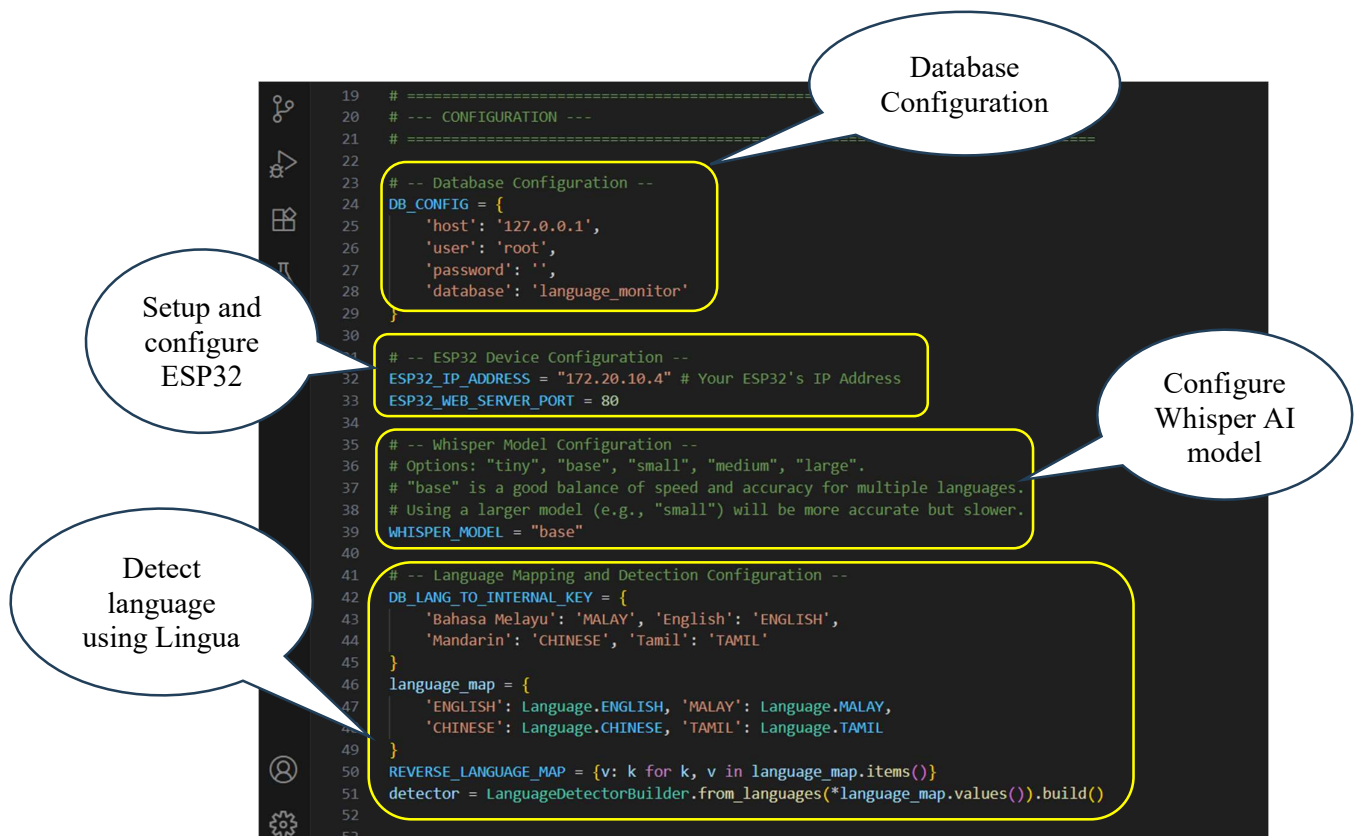


Figure 2-44 Python Section B

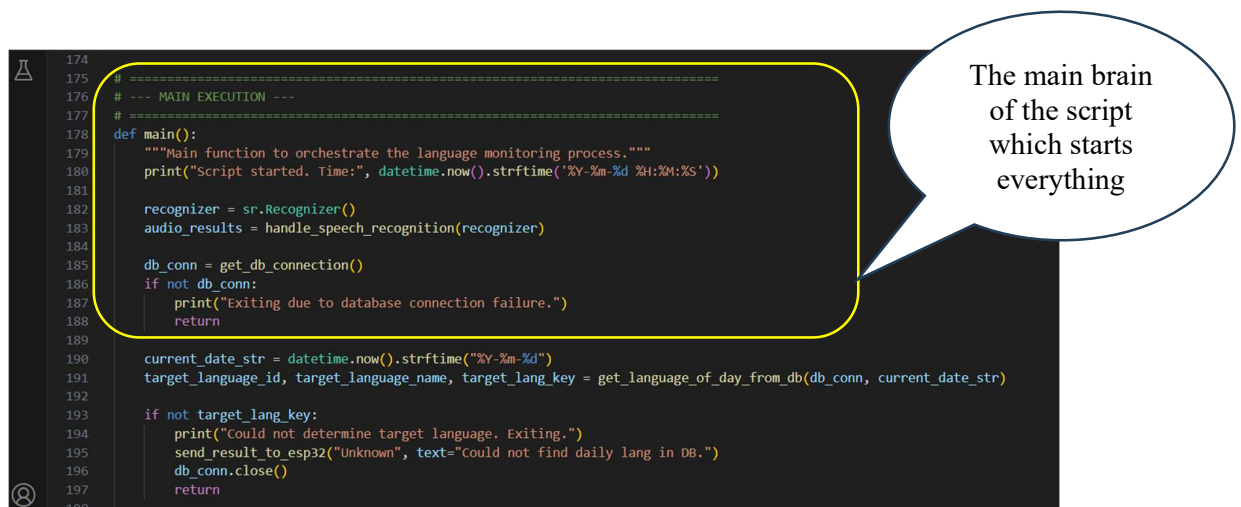


Figure 2-45 Python Section C

These figures above are all small snippets of the big code [which is provided in Appendix A] where these sections play an important role in the whole of the Python source code. As mentioned in the figures, Whisper AI is an open-source speech recognition model developed by OpenAI. This model

can transcribe spoken audio into text in multiple language sand mainly known for its high accuracy even in noisy situations. Since it supports automatic language detection, it helps our system in real-time transcription tasks. Another keyword is Lingua where it is a language detection library that is written in Java but available in Python via bindings. It analyses the text transcribed by the Whisper AI model to determine the language being used whether the sentence is long or short.

b) ESP32 Coding

An ESP32 is a powerful and low-cost microcontroller that is developed by Espressif Systems. Their main goal is to turn an everyday routine task into a smart automatic task. For example, watering the plants can be done automatically with the help of IoT systems and an ESP32 microcontroller by detecting the surrounding temperature and humidity. In our project, it is used as a communication hub and feedback device, i.e., receiving results from a python server and giving real-time audio and display feedback to the user. As mention before in Figure 2-2, most microcontroller such as Arduino, ESP32 and STM32 usually use C or C++ programming language which is tailored for embedded systems. The figures below will show the explanation of a snippet of the coding used in this project via Arduino IDE.

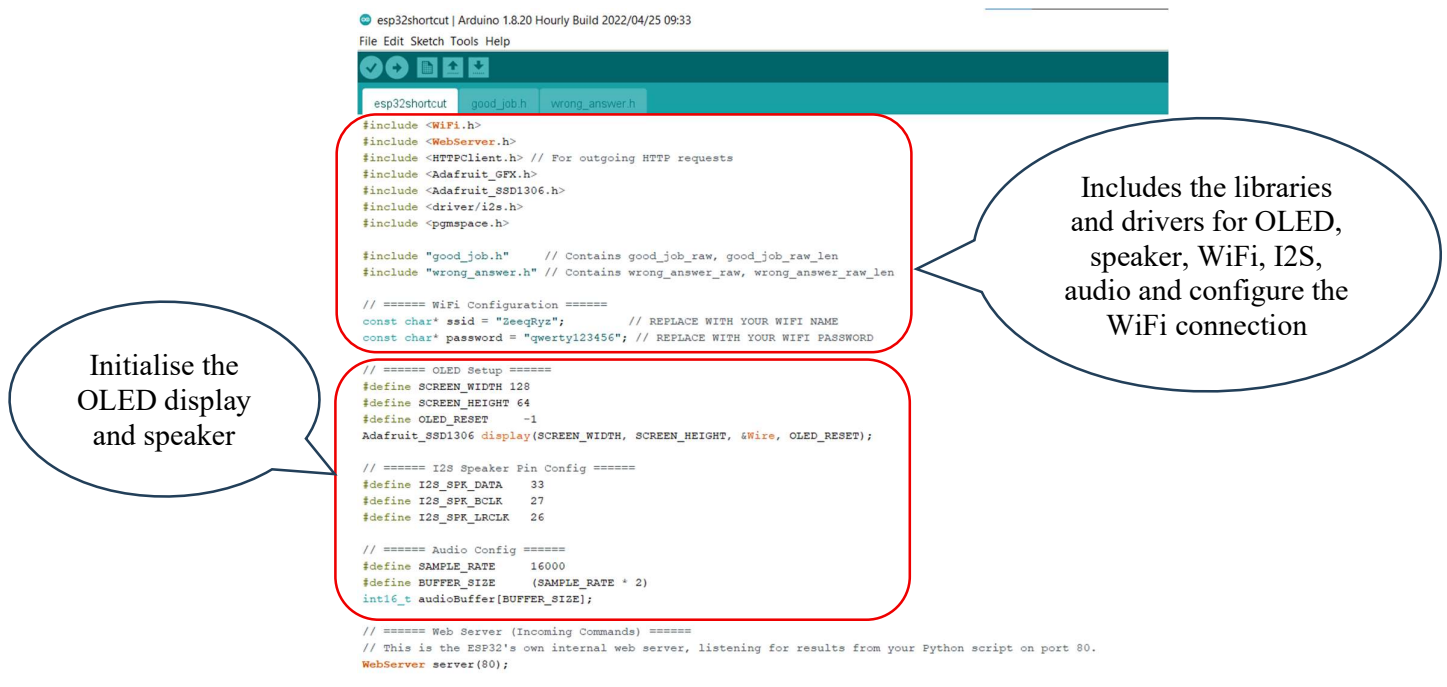


Figure 2-46 ESP32 Code Section A

Handles the incoming commands from python script to ESP32

Figure 2-47 ESP32 Code Section B

[illegible]

Figure 2-48 ESP₃₂ Code Section C

Figure 2-48 shows the header files needed for the speaker to output the phrase “Good Job” and “Wrong Answer, Try Again”. One of the reasons for the usage of header files instead of ordinary audio files is to enable the audio playback without the usage of complex decoding libraries. Furthermore, the process of creating these raw 16-bit PCM data header files begins by using Google Text-To-Speech to generate the spoken phrases (“Good Job” and “Wrong Answer, try again”) as MP3 files. These files were then converted to WAV format using the Pydub library [15], with settings

configured to 16 kHz sample rate, mono channel, and 16-bit sample width which is very much suitable for ESP32 I2S playback. Optionally, Audacity was used to trim the audio or export it directly as RAW (headerless PCM) if needed. Finally, the xxd utility was used to convert the raw audio into a C header file using the command `xxd -i good_job.raw > good_job.h`, producing an array of bytes and its length. These files are included in the ESP32 firmware and stored in flash memory using the PROGMEM keyword, allowing the ESP32 to play them directly through the I2S interface with minimal resource usage.

2.3 Conclusion

All in all, the methodology shows the systematic development of the IoT based smart language monitoring system. From the project design, we can say that the block diagram and hardware used for the system plays a vital role in making this system work and following the objectives given. Next, this circuit provides the clear illustration of what this project was meant to be. For example, the wiring of each component is clearly shown in the schematic making it easy for the creators of this project to create an identical project without any issues. The software implementation also helps the user by getting information about the system usage. By following this methodology, we have managed to create a system that not only includes IoT but also implement AI into this system to make it long-lasting and match it with today's technology progression.

Chapter 3

Results and discussion

3.1 Introduction

This part presents the outcome obtained from the testing of the IoT-based Smart Language Monitoring System and its performance in real-world environments. The findings are compared in terms of accuracy, response time, and reliability, and the anticipated results are contrasted with the actual results. The system's efficiency under diverse environments, such as educational and public service environments, is also compared.

3.2 Results

The system's key features are including real-time language switching, OLED display functionality and voice announcement clarity were tested individually and as an integrated system. The testing also evaluated the user experience and reliability of the system over multiple use cases. This system begins where the student will push the microphone icon on the webpage using the two key keyboard to start listening mode as shown in Figure 3-1.

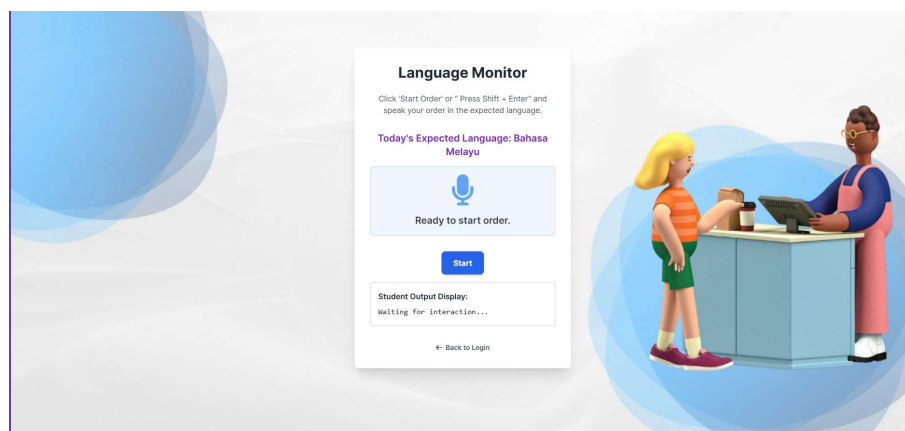


Figure 3-1 Program Start

However, if there is any issue with the microphone or connection to the ESP32 system, an error warning will appear on the webpage itself as shown in Figure 3-2.

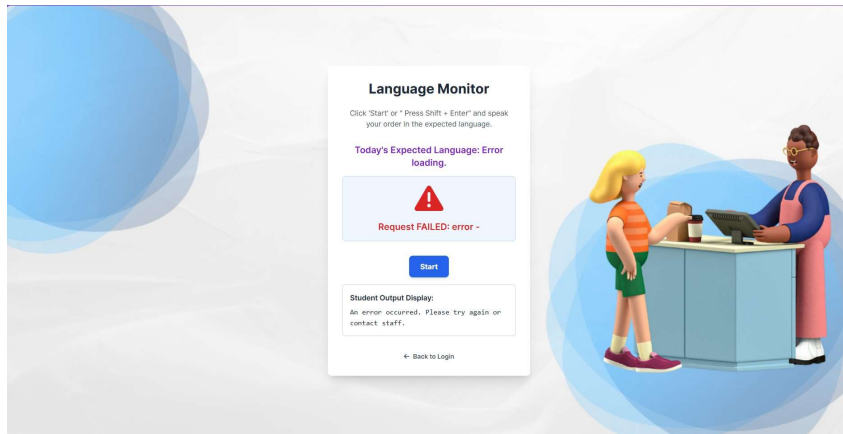


Figure 3-2 Error Warning

3.2.1 Targeted Language: Correct

When the teacher picks the targeted language of the day and the student speaks in the correct language, the OLED screen will display a message that says “Result: CORRECT!”. Furthermore, it will show what the system heard and an audio will be heard from the speaker saying that is the correct answer and the user has done a great job. For example, when the targeted language of the day is Tamil and the student spoke in Tamil (Figure 3-3), the OLED screen will display the message as shown in Figure 3-4.

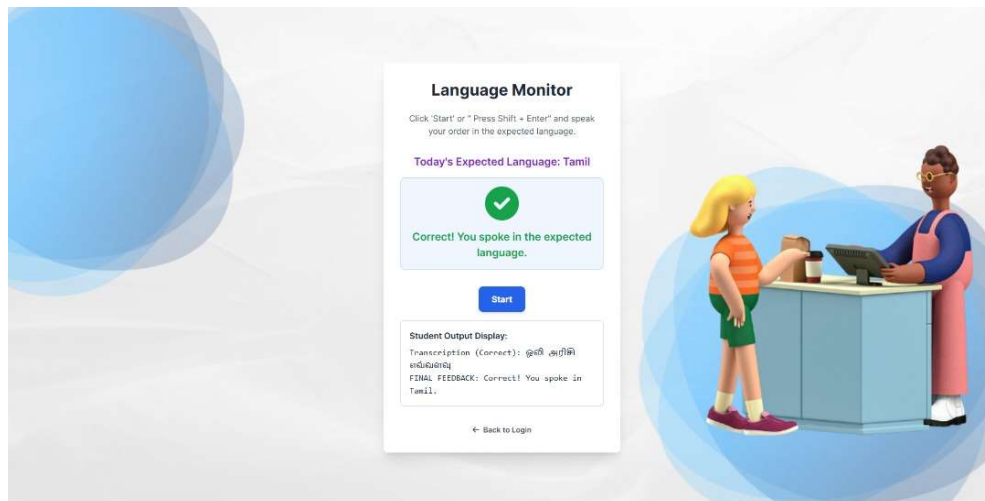


Figure 3-3 Teacher Targeted Language (TAMIL)

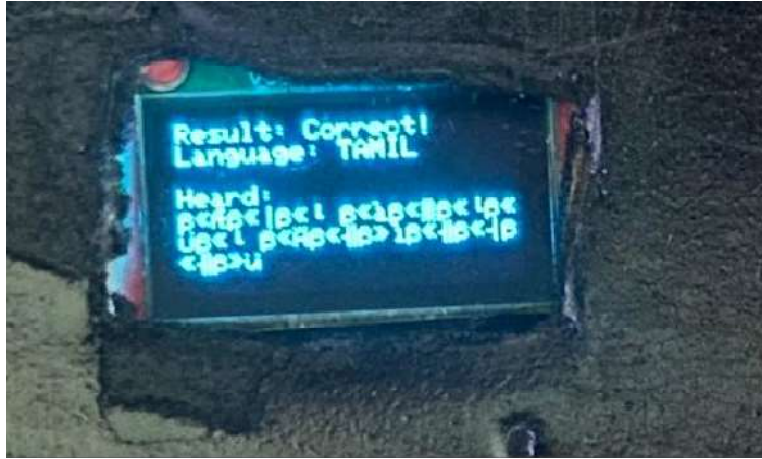


Figure 3-4 Correct OLED Results

3.2.2 Targeted Language: Wrong

If the primary school student does not speak the targeted language, the OLED screen will display a message that says “Result: WRONG LANGUAGE”. Furthermore, it will show what the system heard and what is the targeted language that is needed to be spoken on that day. Then, an audio will be heard from the speaker saying that is the wrong answer and the user should try again. For example, when the targeted language of the day is Malay and the student spoke in English (Figure 3-5), the OLED screen will display the message as shown in Figure 3-6 and the audio can be heard saying that the student should try again.

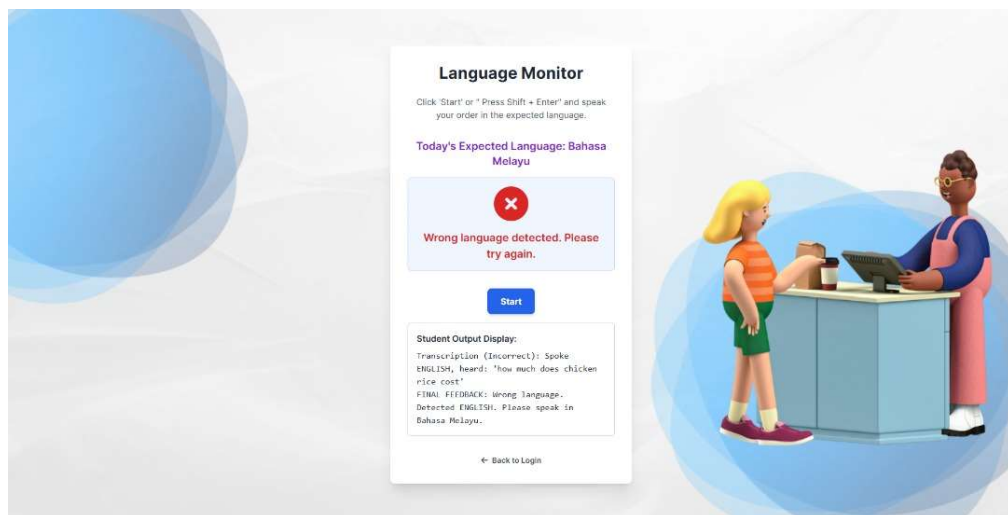


Figure 3-5 Teacher Targeted Language (MALAY)



Figure 3-6 Wrong OLED Results

3.2.3 Dashboard Results



Figure 3-7 Language Usage Overview

The figure above displays the comprehensive Language Usage Overview dashboard accessible to teachers. This page provides visual analytics of student language interactions, including doughnut charts for 'Detected Language Distribution' and 'Correct vs. Incorrect Detections', and a bar chart illustrating 'Weekly Language Usage Trend' from Monday to Friday.

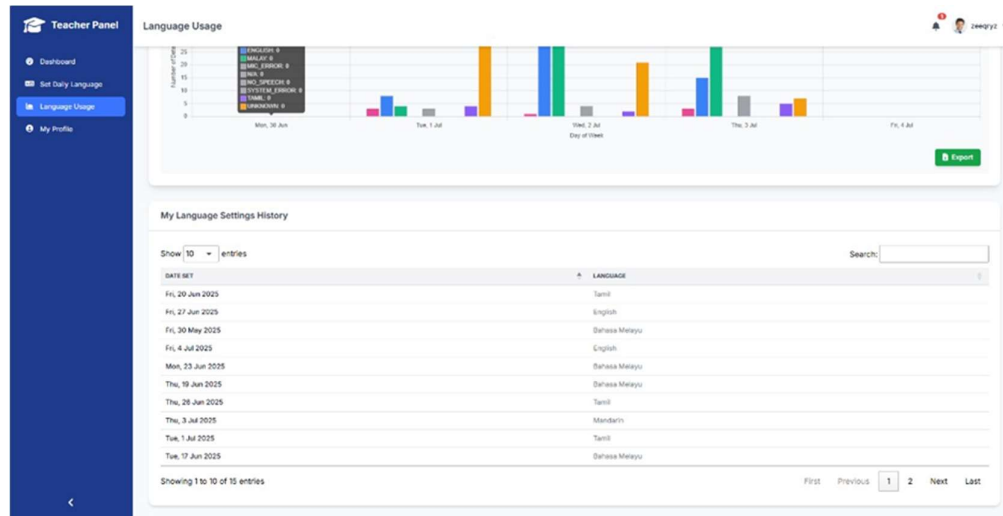


Figure 3-8 Language Settings History

In Figure 3-8 shows the 'My Language Settings History' table within the Language Usage section. This table provides teachers with a chronological log of the daily language settings they have applied, including the date and the selected language, complete with search and pagination features.

3.2.4 Field-testing at SJK Tamil Kangar



Figure 3-9 Meeting with SV and Headmaster



Figure 3-10 Malay Language Zone



Figure 3-12 Testing #1



Figure 3-11 Testing #2

The figures above detail the adventures our team experienced in our field-testing at SJK Tamil Kangar. This school is one of the most prestigious and well-known school located in the heart of Kangar, Perlis. One of the unique traits that this school has is the Malay language Zone located in the cafeteria of the school as shown in Figure 3-10. On the 2nd of July 2025, we arrived at the school around 10a.m which was the beginning of recess time for the primary students. Once arrived, we were met with our supervisor, PM DR. Thennarasan Sabapathy and the headmaster of the school, Mr. Uthaya Kumar Krishnan. We had a short conversation (Figure 3-9) about the prototype and how we are going to produce the results.

After the testing, we begin our testing with one of the students from standard 1, as shown in Figure 3-12, by selecting the Malay language as the “Language of the Day” in our prototype. The results produced from the test were accurate and impressed the student as well. Next, another test was done with a student from Standard 6, as shown in Figure 3-11. In this test, we changed the “Language of the Day” to the English language to determine whether our system can detect and transcribe the spoken language. After the student has ordered their food in English, to our surprise, the system managed to produce another successful result. Overall, our prototype was proven to be a success in our field-testing.

3.3 Conclusion

In general, the IoT-Based Smart Language Monitoring System functioned as expected and completed all the technical and educational objectives outlined at the proposal stage. The primary functions like two-language switching, voice output, and text display were properly integrated and checked. The system was reliable, concise, and user-friendly under a controlled test environment. While the current prototype provides a good proof of concept, there are some rooms for improvement, such as speaker quality, dynamic updating of content, and the inclusion of more languages. Nevertheless, the system has tremendous potential for real-world implementation in primary school canteens and can continue to be improved as a tool for education in multilingual communication and national language policy. For more results of other languages, please refer to the “More Results” section in Appendix A.

Chapter 4

Project Impact and Contribution

4.1 Introduction

This chapter explains how our project, the IoT-Based Smart Language Monitoring System, affects society and the environment. We will discuss both the good and bad points that come from using this system. This section also shows our understanding of what it means to be a professional engineer, including our responsibilities to society, culture, the world, and the environment. Lastly, we will explain our professional and ethical duties and our commitment to the community.

4.2 Health and Safety

For this project, we focused on making sure everything is safe. We thought about safe ways to work and looked at any possible risks. We also made sure to follow high safety standards and regulations:

- **Safe working practices:** We designed the system for use in a primary school cafeteria environment, considering child safety, simplicity, and ease of use.
- **Risk assessment:** We considered potential issues like the need for good internet connectivity and clear voice input, which might be a problem in some schools. The system provides immediate audio-visual feedback, and the results are recorded into a database that can then be easily exported to Excel for teachers.

This process ensures student privacy as no individual voice data is stored. Access to this results database is limited only to authorized teachers.

- **High safety standards and regulations:** As aspiring engineers, we thought about our ethical responsibilities to ensure the system is safe, non-intrusive, and designed in a way that respects the privacy and needs of young users.

4.3 Cultural and Benefit to Society

Our project is carefully designed to fit within community rules, norms, and values. We believe it promotes good values and ethics. It also helps promote societal integration and provides benefits to the society and community.

- **Promotion of good values and ethics:** The project encourages students to use multiple languages actively. This can make them more confident and increase their exposure in a multilingual setting, especially by nurturing the importance of their mother tongue in Malay, while also improving their ability to communicate in English, the international language, which is very relevant in a country like Malaysia.
- **Promotion of societal integration:** From a societal perspective, it promotes inclusivity and cultural respect by rotating languages (Malay, English, Chinese, or Tamil additionally). This gives all major communities a chance to be represented.
- **Benefits to society and community:** This system helps teachers monitor students' language usage effortlessly. The system provides real-time feedback and logs the results into a database, which can be easily exported to Excel. This helps identify language difficulties early and helps students get better at communicating within today's modern technological learning framework. It also helps reduce language barriers, nurture multilingual awareness, and promote equal learning opportunities by making everyday situations part of the learning journey.

4.4 Environment and Sustainability: Sustainable Development Goal 4 - Quality Education

Our project strongly supports the UN Sustainable Development Goal 4: Quality Education. This goal is about promoting inclusive and equitable quality education for all.

- **How the project supports SDG 4:** By using IoT and AI, the system encourages daily language practice at the school canteen counter. It provides students with a fun and real-world opportunity to develop their communication skills specifically in Malay, English, and occasionally Chinese or Tamil during special occasions. This simple interaction helps reinforce classroom learning in a practical setting. Teachers are also empowered with tools to monitor student progress through real-time voice feedback and automated result logging into a database, making it easier to support students even in large classroom environments.¹ In the long run, the system aims to reduce language barriers, nurture multilingual awareness, and promote equal learning opportunities.
- **Environmental footprint:** The project has a low footprint because the components used (ESP32, microphone, OLED, etc.) are energy-efficient. There is also minimal e-waste due to the system's small size.

4.5 Professionalism and ethical responsibilities in project implementation

In carrying out this project, we are committed to following important ethical rules and practices to make sure our work is done responsibly and with honesty. We considered our ethical responsibilities to ensure the system is safe, non-intrusive, and designed in a way that respects the privacy and needs of young users. While the system uses voice recognition to analyze spoken language, it does not store individual student voice data. We ensure that access to the recorded results in the database, which can be exported to Excel, is limited only to authorized teaching staff, and that this data is kept secure. We aimed to create something meaningful, practical, and scalable, especially for communities that often get left behind in digital education efforts.

4.6 Commercialization Potential

Understanding the market and how our project can compete is important.

- **Project Value and Market Impact:** The project addresses a real and urgent educational gap. Teachers expressed strong interest in a tool that could help them identify speech delays, vocabulary usage patterns, and reading fluency, especially for students who may not receive enough language support at home. This tool may also be able to help parents educate their children at home. The system shows promise that it can solve a current issue in most schools: misunderstanding sentences due to a language barrier problem.
- **Targeted Consumers/Market/Users/Applications/Industry:** The primary targeted group are primary school teachers and school administrators from both urban and semi-rural areas. The system is designed for use in a primary school cafeteria environment.
- **Comparison Survey between your product with available product in the market:**

Product	Our IoT-Based Smart Language Monitoring System	Available Product 1: iFLYTEK Smart Translator	Available Product 2: Dexway Language Learning Platform
Cost	Low (Estimated below RM 261.35 for one device)	Varies, typically in the hundreds of USD for a device	Subscription-based, varies by institution size/features
Extra Features or Unique Functions	Real-time audio-visual feedback, Automated result collection into database, Daily language switching in cafeteria, Supports four languages (Malay, English, Chinese, and Tamil), Results exportable to Excel	Real-time face-to-face translation, OCR camera translation, Offline mode, Supports many languages	AI role-play, Virtual assistant, Gamification, Detailed pronunciation feedback (color-coded), Student voice comparison with native speakers, Offline capabilities (app)

Advantages	Encourages multilingual learning in a real-world setting, Improves language proficiency through daily practice, Promotes cultural awareness, Early identification of language difficulties, Reduces teacher monitoring workload with automated data	Portable, Facilitates direct communication between different language speakers, Can be used for quick understanding of foreign text via camera	Comprehensive language courses, Focuses on pronunciation and fluency, Provides detailed and objective feedback, Flexible study options (online/offline), Supports various learning styles
Disadvantages	Relies on good internet connectivity and clear voice input, Limited to basic language recognition functionality	Primarily for translation, not ongoing language usage monitoring or structured assessment in a learning context, Does not collect structured usage data for teachers	Not specifically designed for cafeteria interaction points, May require dedicated learning time outside of natural interactions, Primarily a learning platform rather than a "monitoring system" for daily usage

Table 4-1 Comparison Survey

4.6.1 Project Costing

- **Capital/Material cost**

Materials	Price Per Unit (RM)	Quantity	Total Cost (RM)
ESP32 Developer Module	21.80	1	21.80
Speaker	8.90	1	8.90
I2C OLED Screen	10.00	1	10.00
Micro USB cable	2.00	1	2.00

Microphone	12.30	1	12.30
Jumper Wires	0.5	10	5.00
Total Material Cost			60.00

Table 14-2 Material Cost

- **Operational Cost:** For this project, there would be other operational costs, such as glue to stick the foam board, PVC free, and joiners while assembling the device's outer box. The PVC heating hinge allows the box to be opened and closed.
- **Total Cost (Commercial Price):** The cost estimation for the prototype will be cheaper when we buy all the components for this project. The final price for one device would be lower than RM 70.00.

Chapter 5

Conclusion and Future Work

5.1 Conclusion

This project has been successfully carried out within the time given. All the key objectives that were planned at the beginning have been achieved through careful planning, collaboration, and execution. The system works, and its performance has been evaluated in a simulated canteen environment.

Firstly, we were able to develop a working prototype of the IoT-Based Smart Language Monitoring System that switches languages based on the language that the teacher has picked. This core functionality met our first objective and showed that the system can support bilingual learning in a real-world environment. Secondly, we successfully integrated both display and voice features using hardware like the ESP32, OLED screen, speaker, and microphone. The system gives real-time feedback and stores results into a simple database that can be exported, which achieves our second objective related to usability and monitoring. Additionally, we have managed to integrate an AI voice recognition model to improve and increase our project's functionality. Finally, our project is considered ethical and professional in aspects such as student safety, privacy, and environmental sustainability. We applied safe design practices and ensured that no sensitive data was stored.

The total cost of the prototype remained affordable, showing good potential for real use in schools. In short, the project met all the intended goals, and we believe it contributes meaningfully to both education and technology in schools.

5.2 Future Work

As discussed in Chapter 1, our project focused on developing a basic but, functional prototype targeted for primary school canteens. While the current system meets the core objectives, there are still several areas that can be improved or expanded in the future such as:

- Allow the system to support more languages, especially Chinese and Tamil, to reflect Malaysia's multilingual environment. This would make the system more inclusive and relevant to a wider range of primary schools.
- To develop a user-friendly web interface that lets teachers update the phrases, schedule language settings, and check the database without having to access the hardware directly.
- Improve the voice recognition function with clearer audio processing and support for offline mode would help schools that have limited internet access.
- Add basic analytics to track language use and generate simple reports which can help teachers identify students' language performance more effectively.
- The physical design of the device can be improved by using a more professional enclosure, possibly using 3D printing or laser-cut acrylic, to make it more durable and suitable for long-term use in real school environments.
- Make it completely wireless by adding a better microcontroller to hopefully run everything on it including the Google AI speech recognition processing instead of making it run on PC.

In conclusion, this project has good potential to grow further. With additional development, it could become a valuable educational tool not only for schools in Malaysia but also for multilingual learning environments in other countries.

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APPENDIX A

GANTT CHART

Activities	Week													
	1	2	3	4	5	6	7	8	9	10	11	12	13	14
IDP TALK	/													
PROJECT TITLE SELECTION AND PLANNING		/	/											
DOING PROJECT SYNOPSIS				/										
COMPONENT SELECTION AND HARDWARE PLANING					/									
SOFTWARE DESIGN AND LOGIC DEVELOPMENT						/	/							
LANGUAGE SWITCHING LOGIC								/						
IOT MODULE CONFIGURATION AND TESTING									/					
SYSTEM INTEGRATION AND TROUBLESHOOTING										/				
FINAL TESTING AND PERFORMANCE ANALYSIS											/			
REPORT WRITING AND DOCUMENTATION												/	/	
FINAL VIVA PRESENTATION														/

ESP32 SOURCE CODE

```

#include <WiFi.h>
#include <WebServer.h>
#include <HTTPClient.h> // For outgoing HTTP requests
#include <Adafruit_GFX.h>
#include <Adafruit_SSD1306.h>
#include <driver/i2s.h>
#include <pgmspace.h>

```

```

#include "good_job.h" // Contains good_job_raw, good_job_raw_len
#include "wrong_answer.h" // Contains wrong_answer_raw,
wrong_answer_raw_len

// ===== WiFi Configuration =====
const char* ssid = "ZeeqRyz"; // REPLACE WITH YOUR WIFI NAME
const char* password = "ayamgepuk"; // REPLACE WITH YOUR WIFI
PASSWORD

// ===== OLED Setup =====
#define SCREEN_WIDTH 128
#define SCREEN_HEIGHT 64
#define OLED_RESET -1
Adafruit_SSD1306 display(SCREEN_WIDTH, SCREEN_HEIGHT, &Wire,
OLED_RESET);

// ===== I2S Speaker Pin Config =====
#define I2S_SPK_DATA 33
#define I2S_SPK_BCLK 27
#define I2S_SPK_LRCLK 26

// ===== Audio Config =====
#define SAMPLE_RATE 16000

// ===== Web Server (Incoming Commands) =====
WebServer server(80);

// ===== Server Details (Outgoing Requests from ESP32 to PC) =====
const char* phpHandlerUrl =
"http://172.20.10.2/esp_communication_handler.php"; // IP of your PC
running XAMPP

```

```

const char* deviceID = "ESP32_LangMon_002";    // Unique ID for this
device.

const char* apiKey = "ESP32_SECRET_KEY"; // !!! IMPORTANT: MUST
MATCH API_KEY IN YOUR PHP SCRIPT !!!

// ===== Update Interval (Outgoing Status) =====
unsigned long lastStatusUpdateTime = 0;
const unsigned long statusUpdateInterval = 60000; // Send status every 60
seconds

// ===== OLED Display Function =====
void showOnOLED(String line1, String line2 = "") {
    display.clearDisplay();
    display.setTextSize(1);
    display.setTextColor(SSD1306_WHITE);
    display.setCursor(0, 10);
    display.println(line1);
    if (line2 != "") {
        display.setCursor(0, 30);
        display.println(line2);
    }
    display.display();
}

// ===== Function to display detailed results on the OLED =====
void showDetailedResultOnOLED(String result, String text, String detected,
String expected) {
    display.clearDisplay();
    display.setTextSize(1);
    display.setTextColor(SSD1306_WHITE);
    display.setCursor(0, 0);
    display.setTextWrap(true); // Enable text wrapping for long lines

```

```

if (result.equalsIgnoreCase("Correct")) {
    display.println("Result: Correct!");
    display.println("Language: " + detected);
    display.println("");
    display.println("Heard:");
    display.print(text);
} else if (result.equalsIgnoreCase("Wrong")) {
    display.println("Result: WRONG LANG");
    display.println("Heard: " + detected);
    display.println("Need: " + expected);
    display.println("");
    display.print("Text:" + text);
} else {
    // Handle other errors like NoSpeech, MicError etc.
    showOnOLED("Error", result);
}
display.display();
}

// ===== Audio Playback Function =====
void playAudio(const unsigned char* rawData, size_t len) {
    size_t bytes_written;
    size_t samples = len / 2;

    i2s_zero_dma_buffer(I2S_NUM_0);

    for (size_t i = 0; i < samples; i++) {
        int16_t sample = (int16_t)(pgm_read_byte_near(rawData + 2 * i) |
                                   (pgm_read_byte_near(rawData + 2 * i + 1) << 8));
        i2s_write(I2S_NUM_0, (const char*)&sample, sizeof(sample),
        &bytes_written, portMAX_DELAY);
    }
}

```

```

Serial.println("[AUDIO] Playback finished.");
}

// ===== I2S Setup =====
void setupI2SSpeaker() {
  i2s_config_t i2s_config = {
    .mode = (i2s_mode_t)(I2S_MODE_MASTER | I2S_MODE_TX),
    .sample_rate = SAMPLE_RATE,
    .bits_per_sample = I2S_BITS_PER_SAMPLE_16BIT,
    .channel_format = I2S_CHANNEL_FMT_ONLY_RIGHT,
    .communication_format = I2S_COMM_FORMAT_I2S_MSB,
    .intr_alloc_flags = 0,
    .dma_buf_count = 8,
    .dma_buf_len = 1024,
    .use_apll = false,
    .tx_desc_auto_clear = true,
    .fixed_mclk = 0
  };

  i2s_pin_config_t pin_config = {
    .bck_io_num = I2S_SPK_BCLK,
    .ws_io_num = I2S_SPK_LRCLK,
    .data_out_num = I2S_SPK_DATA,
    .data_in_num = I2S_PIN_NO_CHANGE
  };

  i2s_driver_install(I2S_NUM_0, &i2s_config, 0, NULL);
  i2s_set_pin(I2S_NUM_0, &pin_config);
  i2s_zero_dma_buffer(I2S_NUM_0);
  Serial.println("[I2S] Speaker setup complete.");
}

// ===== WiFi Connection Function =====

```

```

void connectToWiFi() {
  if (WiFi.status() == WL_CONNECTED) {
    return;
  }
  Serial.print("Connecting to WiFi: ");
  Serial.println(ssid);
  showOnOLED("Connecting to", ssid);

  WiFi.begin(ssid, password);
  int attemptCount = 0;
  while (WiFi.status() != WL_CONNECTED && attemptCount < 30) {
    delay(500);
    Serial.print(".");
    showOnOLED("Connecting.", String(attemptCount));
    attemptCount++;
  }

  if (WiFi.status() == WL_CONNECTED) {
    Serial.println("\n✅ WiFi Connected!");
    IPAddress ip = WiFi.localIP();
    Serial.print("IP address: ");
    Serial.println(ip);
    showOnOLED("Connected!", ip.toString());
  } else {
    Serial.println("\n❌ Failed to connect to WiFi. Retrying...");
    showOnOLED("WiFi Failed!", "Retrying...");
  }
}

// ===== Send Status Function (Outgoing HTTP POST to PC Server)
=====

void sendStatus(String currentStatus) {
  if (WiFi.status() == WL_CONNECTED) {

```

```

    HTTPClient http;

    String url = String/phpHandlerUrl) + "?action=update_status";

    if (http.begin(url)) {
        http.addHeader("Content-Type", "application/x-www-form-
urlencoded");

        String postData = "api_key=" + String(apiKey) +
            "&device_id=" + String(deviceID) +
            "&status=" + currentStatus;

        int httpCode = http.POST(postData);
        if (httpCode > 0) {
            Serial.printf("[HTTP] POST... code: %d\n", httpCode);
        } else {
            Serial.printf("[HTTP] POST failed, error: %s\n",
http.errorToString(httpCode).c_str());
        }
        http.end();
    } else {
        Serial.printf("[HTTP] Unable to begin connection to server: %s\n",
url.c_str());
    }
    } else {
        Serial.println("WiFi Disconnected. Cannot send status.");
    }
}

// ===== HTTP Handler (Incoming Commands from Python Script to
ESP32) =====
void handleRoot() {
    if (server.hasArg("result")) {
        // Read all the arguments from the URL
        String result = server.arg("result");
    }
}

```

```

String text = server.hasArg("text") ? server.arg("text") : "N/A";
String detected = server.hasArg("detected") ? server.arg("detected") :
"N/A";
String expected = server.hasArg("expected") ? server.arg("expected") :
"N/A";

// Print to Serial Monitor for debugging
Serial.println("📡 Received detailed result from Python Script.");
Serial.println("Result: " + result);
Serial.println("Text: " + text);
Serial.println("Detected: " + detected);
Serial.println("Expected: " + expected);

// Call the new display function to show details on the OLED
showDetailedResultOnOLED(result, text, detected, expected);

// Play the corresponding audio feedback
if (result.equalsIgnoreCase("Correct")) {
    playAudio(good_job_raw, good_job_raw_len);
} else if (result.equalsIgnoreCase("Wrong")) {
    playAudio(wrong_answer_raw, wrong_answer_raw_len);
}

// Send a confirmation response back to the Python script
server.send(200, "text/plain", "Detailed result received: " + result);

} else {
    server.send(200, "text/plain", "No 'result' arg provided.");
    showOnOLED("Waiting for", "Trigger...");
}
}

```

```

// ===== Setup Function =====
void setup() {
  Serial.begin(115200);
  delay(100);
  Serial.println("\nESP32 Language Monitor - Starting Up");

  // Initialize OLED display
  if (!display.begin(SSD1306_SWITCHCAPVCC, 0x3C)) {
    Serial.println("❌ OLED init failed!");
    while (true) { delay(100); }
  }
  showOnOLED("Starting Up...", "Please Wait");

  // Connect to WiFi network
  connectToWiFi();

  // Setup I2S speaker
  setupI2SSpeaker();

  // Start ESP32's internal web server to listen for incoming commands from
  Python
  server.on("/", handleRoot);
  server.begin();
  Serial.println("🚀 HTTP Server (for incoming commands) Started on port
80.");

  // Send initial "online" status to the PC server if WiFi is connected
  if (WiFi.status() == WL_CONNECTED) {
    sendStatus("online");
    lastStatusUpdateTime = millis();
  }
}

```

```

// ===== Loop Function =====
void loop() {
  server.handleClient();

  unsigned long currentTime = millis();
  if (WiFi.status() == WL_CONNECTED) {
    if (currentTime - lastStatusUpdateTime >= statusUpdateInterval) {
      sendStatus("online");
      lastStatusUpdateTime = currentTime;
    }
  } else {
    Serial.println("Wi-Fi disconnected. Attempting to reconnect...");
    showOnOLED("WiFi Lost!", "Reconnecting...");
    connectToWiFi();
    delay(1000);
  }
}

```

AI / PYTHON SOURCE CODE

```
import speech_recognition as sr
from gtts import gTTS
from playsound import playsound
from datetime import datetime
from lingua import Language, LanguageDetectorBuilder
import mysql.connector
import os
import requests
import sys
import urllib.parse
import numpy as np
import noisereducer as nr
import io

# --- Enforce UTF-8 for all print output ---
sys.stdout = io.TextIOWrapper(sys.stdout.buffer, encoding='utf-8')
sys.stderr = io.TextIOWrapper(sys.stderr.buffer, encoding='utf-8')

#
=====

=====

# --- CONFIGURATION ---
#
=====

=====

# -- Database Configuration --
DB_CONFIG = {
    'host': '127.0.0.1',
    'user': 'root',
```

```

    'password': '',
    'database': 'language_monitor'
}

# -- ESP32 Device Configuration --
ESP32_IP_ADDRESS = "172.20.10.4" # Your ESP32's IP Address
ESP32_WEB_SERVER_PORT = 80

# -- Whisper Model Configuration --
# Options: "tiny", "base", "small", "medium", "large".
# "base" is a good balance of speed and accuracy for multiple languages.
# Using a larger model (e.g., "small") will be more accurate but slower.
WHISPER_MODEL = "base"

# -- Language Mapping and Detection Configuration --
DB_LANG_TO_INTERNAL_KEY = {
    'Bahasa Melayu': 'MALAY', 'English': 'ENGLISH',
    'Mandarin': 'CHINESE', 'Tamil': 'TAMIL'
}
language_map = {
    'ENGLISH': Language.ENGLISH, 'MALAY': Language.MALAY,
    'CHINESE': Language.CHINESE, 'TAMIL': Language.TAMIL
}
REVERSE_LANGUAGE_MAP = {v: k for k, v in language_map.items()}
detector =
LanguageDetectorBuilder.from_languages(*language_map.values()).build()

#
=====

=====

# --- HELPER FUNCTIONS (No changes needed here) ---

```

```

#
=====

def play_feedback_sound(text, lang='en'):
    """Generates and plays an audio feedback, then cleans up the audio
    file."""
    try:
        tts = gTTS(text=text, lang=lang)
        audio_file = "feedback.mp3"
        tts.save(audio_file)
        # playsound(audio_file) # Uncomment for server-side audio feedback
    except Exception as e:
        print(f"Error playing sound: {e}")
    finally:
        if os.path.exists(audio_file):
            os.remove(audio_file)

def get_db_connection():
    """Establishes and returns a database connection."""
    try:
        return mysql.connector.connect(**DB_CONFIG)
    except mysql.connector.Error as err:
        print(f"DB ERROR: Error connecting to database: {err}")
        return None

def get_language_of_day_from_db(conn, date_str):
    """Fetches the global language set for a given date from the database."""
    if not conn: return None, None, None
    with conn.cursor(dictionary=True) as cursor:
        query = "SELECT tdl.language_id, l.language_name FROM
teacher_daily_languages tdl JOIN languages l ON tdl.language_id = l.id
WHERE tdl.setting_date = %s LIMIT 1"

```

```

        cursor.execute(query, (date_str,))
        result = cursor.fetchone()
        if result:
            internal_key =
DB_LANG_TO_INTERNAL_KEY.get(result['language_name'])
            return result['language_id'], result['language_name'], internal_key
        return None, None, None

def log_to_db(query, params, conn):
    """Generic function to execute a logging query to the database."""
    if not conn: return
    try:
        with conn.cursor() as cursor:
            cursor.execute(query, params)
            conn.commit()
    except mysql.connector.Error as err:
        print(f"DB ERROR: Failed to log: {err}")
        conn.rollback()

def send_result_to_esp32(result_status, text="N/A", detected="N/A",
expected="N/A"):
    """Sends the detailed detection result as an HTTP GET request to the
ESP32."""
    try:
        encoded_text = urllib.parse.quote(text)
        url =
f"http://{ESP32_IP_ADDRESS}:{ESP32_WEB_SERVER_PORT}/?result={r
result_status}&text={encoded_text}&detected={detected}&expected={expected
}"
        print(f"[HTTP] Sending detailed result to ESP32: {url}")
        response = requests.get(url, timeout=5)
        print(f"Sent result '{result_status}' to ESP32. ESP32 Response:
{response.status_code} - {response.text}")

```

```

except requests.exceptions.RequestException as e:
    print(f"ERROR: Could not send result to ESP32 at
{ESP32_IP_ADDRESS}: {e}")

#
=====

=====

# --- CORE SPEECH PROCESSING FUNCTION (Updated for Whisper) ---
#
=====

=====

def handle_speech_recognition(recognizer):
    """
    Listens for audio, processes it with Whisper, and returns the results.
    """
    results = { "transcribed_text": "N/A", "detected_lang_key": "N/A",
"error": None }

    # 1. Capture and clean audio
    try:
        with sr.Microphone(sample_rate=16000) as source:
            #print("Calibrating for ambient noise...")
            recognizer.adjust_for_ambient_noise(source, duration=1)
            print("Listening...")
            audio_data = recognizer.listen(source, timeout=5,
phrase_time_limit=15)
            print("Audio captured. Now cleaning noise...")
            audio_data_np = np.frombuffer(audio_data.get_raw_data(),
dtype=np.int16)
            reduced_noise_audio = nr.reduce_noise(y=audio_data_np,
sr=source.SAMPLE_RATE, prop_decrease=0.8)

```

```

        cleaned_audio = sr.AudioData(reduced_noise_audio.tobytes(),
source.SAMPLE_RATE, source.SAMPLE_WIDTH)
        print("Noise cleaning complete. Processing...")

except sr.WaitTimeoutError:
    results["error"] = "NoSpeech"
    return results
except Exception as e:
    results["error"] = "MicError"
    results["transcribed_text"] = f"MIC_ERROR: {e}"
    return results

# 2. Transcribe using Whisper and then detect language using Lingua
try:
    print(f"Processing audio with Whisper (Model:
{WHISPER_MODEL})...")
    # Let Whisper auto-detect the language by setting language=None
    transcribed_text = recognizer.recognize_whisper(cleaned_audio,
model=WHISPER_MODEL, language=None)
    results["transcribed_text"] = transcribed_text.strip()

    if not results["transcribed_text"]:
        # Whisper returned an empty string, treat as not understood
        print("Whisper could not understand the audio.")
        results["error"] = "Unknown"
    else:
        print(f"Whisper transcribed: '{results['transcribed_text']}'")
        # Use Lingua for a final, reliable language check on the text
        detected_lingua_lang =
detector.detect_language_of(results["transcribed_text"])
        if detected_lingua_lang:
            results["detected_lang_key"] =
REVERSE_LANGUAGE_MAP.get(detected_lingua_lang, "UNKNOWN")

```

```

        else:
            results["detected_lang_key"] = "UNKNOWN"
            print(f'Lingua detected language of text as:
{results[\'detected_lang_key\']}')

    except sr.UnknownValueError:
        print("Whisper could not understand the audio.")
        results["error"] = "Unknown"
    except Exception as e:
        print(f'An unexpected error occurred with Whisper: {e}')
        results["error"] = "SystemError"
        results["transcribed_text"] = f"WHISPER_ERROR: {e}"

    return results

#
=====

=====
# --- MAIN EXECUTION ---
#
=====

=====

def main():
    """Main function to orchestrate the language monitoring process."""
    print("Script started. Time:", datetime.now().strftime('%Y-%m-%d
%H:%M:%S'))

    recognizer = sr.Recognizer()
    audio_results = handle_speech_recognition(recognizer)

    db_conn = get_db_connection()
    if not db_conn:
        print("Exiting due to database connection failure.")

```

```

    return

    current_date_str = datetime.now().strftime("%Y-%m-%d")
    target_language_id, target_language_name, target_lang_key =
get_language_of_day_from_db(db_conn, current_date_str)

    if not target_lang_key:
        print("Could not determine target language. Exiting.")
        send_result_to_esp32("Unknown", text="Could not find daily lang in
DB.")
        db_conn.close()
        return

    final_result = "Wrong"
    final_result_for_esp32 = "Unknown"
    feedback_msg = ""

    if audio_results["error"]:
        final_result_for_esp32 = audio_results["error"]
        if final_result_for_esp32 == "NoSpeech":
            feedback_msg = "Sorry, I didn't hear anything. Please try again."
        elif final_result_for_esp32 == "Unknown":
            feedback_msg = f"Sorry, I couldn't understand what you said. Please
try speaking in {target_language_name}."
        else: # MicError, SystemError
            feedback_msg = "An error occurred. Please check the console."
    elif audio_results["detected_lang_key"] == target_lang_key:
        final_result, final_result_for_esp32 = "Correct", "Correct"
        feedback_msg = f"Correct! You spoke in {target_language_name}."
        print(f"Transcription (Correct): {audio_results['transcribed_text']}")
    else: # Language was detected, but it was the wrong one
        final_result, final_result_for_esp32 = "Wrong", "Wrong"

```

```

        feedback_msg = f"Wrong language. Detected
{audio_results['detected_lang_key']}. Please speak in
{target_language_name}."

        print(f"Transcription (Incorrect): Spoke
{audio_results['detected_lang_key']}, heard:
'{audio_results['transcribed_text']}'")

    # Log results to the database
    now = datetime.now()

    log_to_db("INSERT INTO student_interaction_logs (log_date, log_time,
transcribed_text, detected_language, expected_language, result_status)
VALUES (%s, %s, %s, %s, %s, %s)",
            (now.strftime("%Y-%m-%d"), now.strftime("%H:%M:%S"),
audio_results["transcribed_text"], audio_results["detected_lang_key"],
target_lang_key, final_result_for_esp32), db_conn)

    status_enum = 'correct' if final_result == 'Correct' else 'incorrect'
    log_to_db("INSERT INTO language_usage (usage_date,
detected_language, status, language_id) VALUES (%s, %s, %s, %s)",
            (now.strftime("%Y-%m-%d"), audio_results["detected_lang_key"],
status_enum, target_language_id), db_conn)

    # Provide final feedback (sound and ESP32)
    if feedback_msg:
        print(f"\nFINAL FEEDBACK: {feedback_msg}")
        play_feedback_sound(feedback_msg, lang='en')
        send_result_to_esp32(
            final_result_for_esp32,
            audio_results['transcribed_text'],
            audio_results['detected_lang_key'],
            target_lang_key
        )

```

```
db_conn.close()
print("\nScript finished.")
```

```
if _name_ == "_main_":
    main()
```

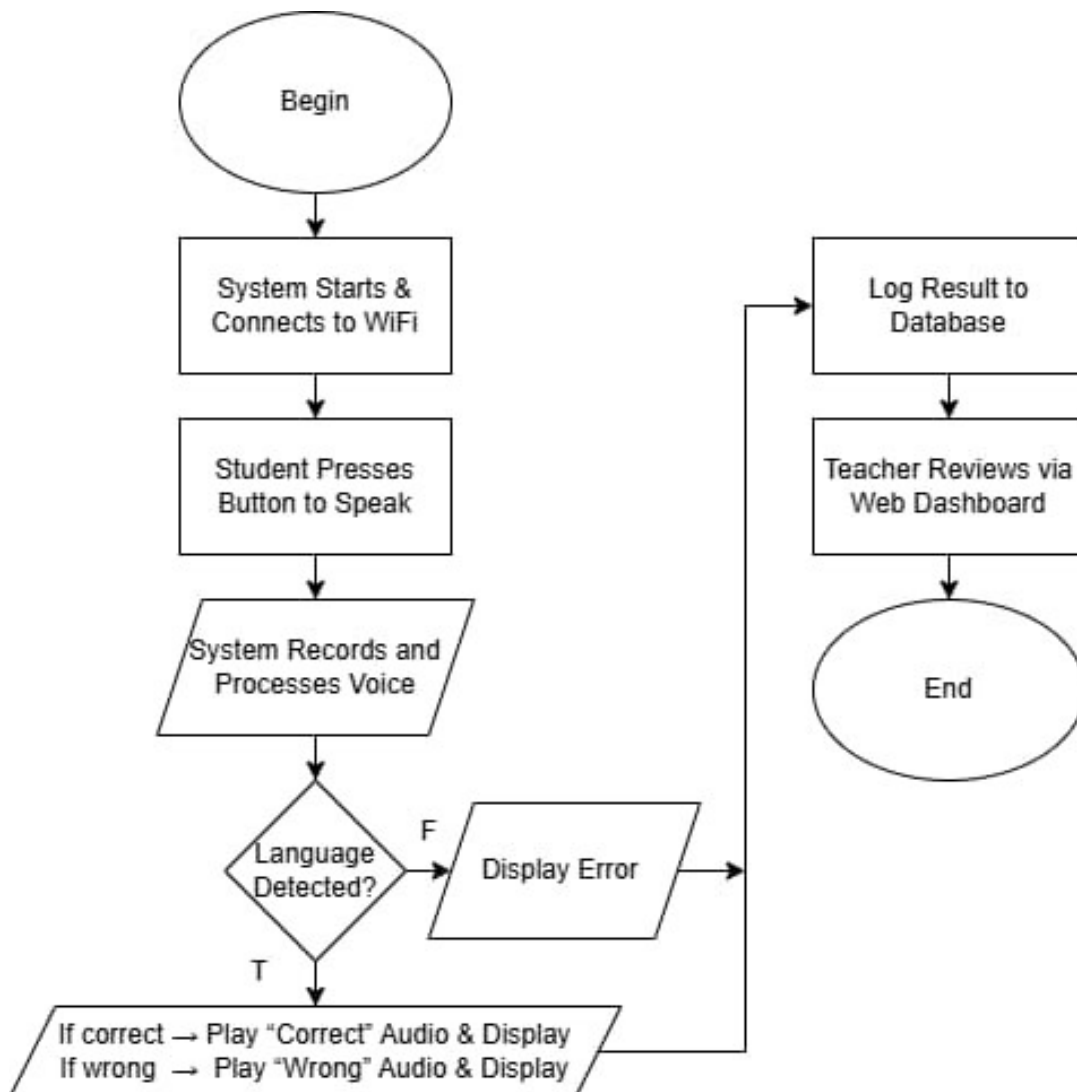


Figure 5-1 Overall system Flowchart

MORE RESULTS

1) Targeted Language : ENGLISH

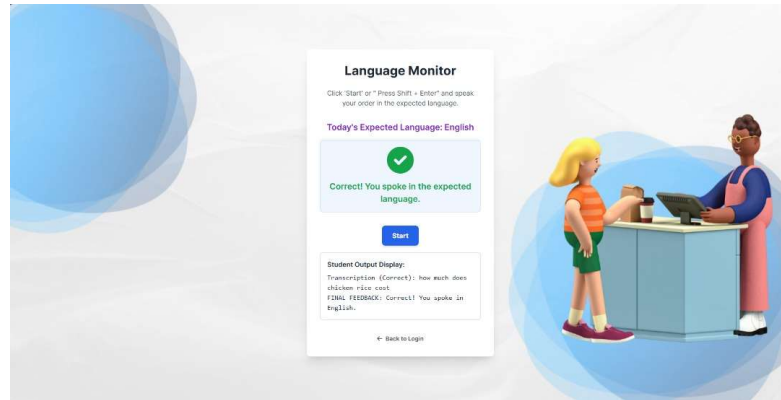


Figure 5-2 ENGLISH CORRECT

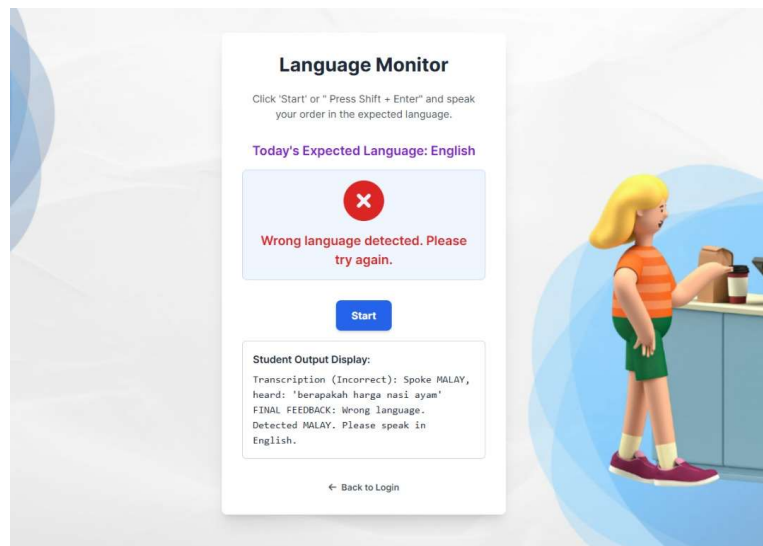


Figure 5-3 ENGLISH WRONG



Figure 5-4 ENGLISH OLED WRONG



Figure 5-5 ENGLISH OLED CORRECT

2) Targeted Language : CHINESE

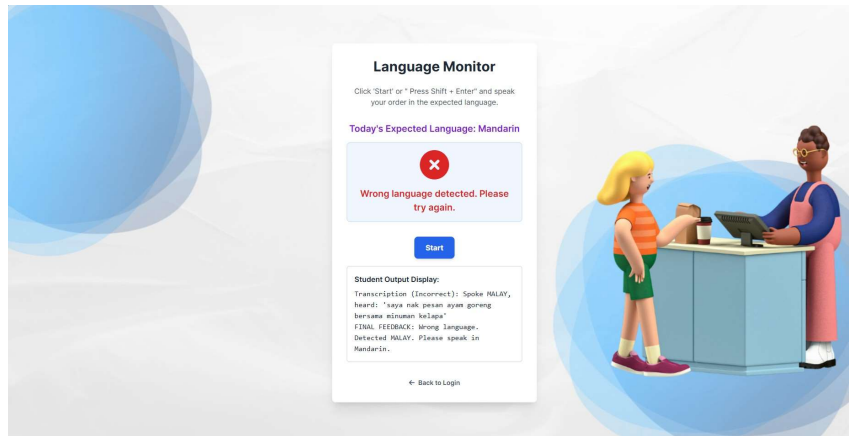


Figure 5-9 CHINESE WRONG

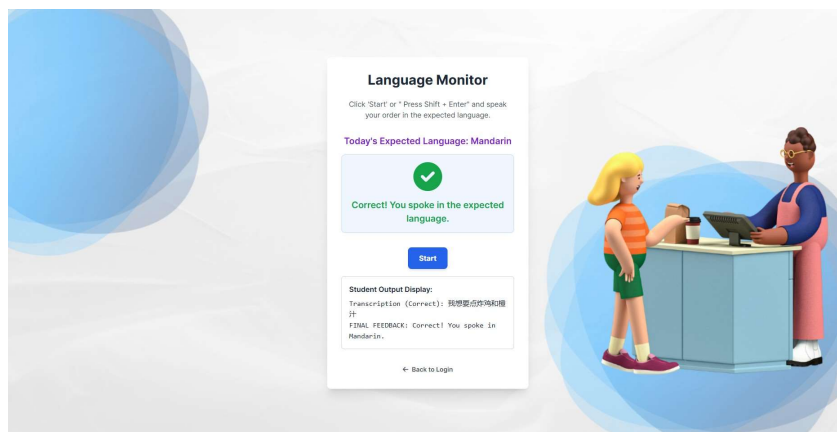


Figure 5-8 CHINESE CORRECT



Figure 5-7 CHINESE OLED WRONG



Figure 5-6 CHINESE OLED CORRECT

APPENDIX B

LIST OF SDG

SDG and Goal Title	Why the Group Chose This SDG	Implimentation to the Project
SDG 4: Quality Education	To ensure inclusive and equitable education and promote lifelong learning opportunities for all.	The system teaches students from a young age to start learning and talking different languages in their daily life. For example, this project can educate students to speak different languages when ordering food in their canteen.
SDG 9: Industry, Innovation, and Infrastructure	To build and foster innovation as well as upgrade the infrastructure to ensure equal and universal access to information.	The system uses AI and IoT to upgrade the traditional speech recognition system in a budget-friendly way For example, This project uses AI to detect various types of languages spoken based on different phrases and words of their respective language and not specific phrases.
SGD 10: Reduced Inequalities	To promote different languages to many people from different lifestyles regardless of gender, race, religious beliefs or economic status.	This system teaches primary students from different lifestyles to learn various languages. For example, the teacher is able to add in more languages to promote more races and lifestyles outside of Malaysia's multiracial society.
SGD 16: Peace, Justice and Strong Institutions	To help ensure fair communication, language transparency and inclusivity in multilingual societies.	This system can be used in a government setting or legal setting to ensure fairness in a multilingual society. For example, it can also be used in places like airports and immigration offices to help foreigners identify the languages being spoken.